

Chance constrained optimal power-water flow: An iterative algorithm assisted by deep neural networks

Xia Zhao^{*}, Dingyi Liu, Zhenyu Wu, Chongqi Huang

National Key Laboratory of Power Transmission Equipment Technology (Chongqing University), Chongqing 400044, China

ARTICLE INFO

Keywords:

Water-energy nexus
Optimal power flow
Chance constraint programming
Neural network
Uncertainty

ABSTRACT

The optimal power-water flow (OPWF) is a fundamental tool for operating integrated power and water networks. To address uncertainties in renewable energy and load demands, researchers have used chance-constrained models to investigate OPWF under uncertainty. However, existing work has limitations, such as over-simplified network representations, intractable problem formulations, and high computational burdens. This paper presents a chance-constrained OPWF formulation that incorporates detailed network constraints and stochastic uncertainties from wind speeds and electricity/water demands. By leveraging the concept of quantiles of stochastic variables and convexification techniques, the model is recast into a deterministic mixed-integer second-order cone programming (MISOCP) problem with uncertain margins. To efficiently solve this problem, we propose a novel iterative algorithm assisted by deep neural networks (DNNs). The algorithm integrates two DNNs: one designed to solve power and water flow equations, accelerating the probabilistic evaluation of uncertain margins; and the other to predict pipeline flows and refine variable ranges, facilitating a warm-start approach to the MISOCP model. Case studies conducted on the IEEE 123-node feeder coupled with a 67-node water distribution network validate the two DNNs and the overall iterative algorithm. The findings demonstrate that the proposed DNN-assisted approach effectively balances system security and economy. Compared to existing algorithms, the relative error of the optimization solution is less than 0.03%, with a speedup ratio exceeding 200.

1. Introduction

1.1. Motivations

The electrical power and water systems are typically planned and operated as separate entities, although they are intrinsically interdependent. This interdependency, known as the *water-energy nexus* [1,2], has become increasingly significant due to the growing demand for both electricity and water resources. It is estimated that approximately 40 % of the water withdrawn is used for power production, specifically for cooling and fuel extraction [1]. On the other hand, water and wastewater services are highly energy-intensive, accounting for about 3–4 % of the total electricity consumption in the USA [3], and the average energy consumption for urban water supply in China in 2011 was recorded at 0.29 kWh/m³ [4]. In addition, over 80 % of the energy consumption within the water sector is ascribed to water-pumping activities [2].

Over the last decade, global attention has intensified on the

development of *multi-energy systems* [5,6]. This evolution has led to the emergence of diverse multi-energy configurations, including power-gas [7,8], power-heat [9], power-transportation [10], power-gas-heat [11], and power-gas-hydrogen systems [12]. The goal of these integrated systems is to identify synergies between different energy carriers, enhancing overall efficiency, reliability, and sustainability. Recently, there has been growing interest in exploring synergies between power and water systems, aligning with the broader concept of multi-energy systems. Numerous studies have proposed the coordinated operation of these two systems from an integrated perspective [3,13–16], with a specific problem known as the *optimal power-water flow* (referred to as optimal PWF or OPWF, and also named *optimal water-power flow* in [13]).

The OPWF problem is an extension of the classic optimal power flow (OPF) problem for electrical power systems within the context of integrated power and water systems (IPWSs). It aids system operators in making well-informed decisions to minimize the overall operational costs, meet power and water demands, and satisfy engineering and

^{*} Corresponding author at: Institute: National Key Laboratory of Power Transmission Equipment Technology, Address: Chongqing University, Chongqing, China.
E-mail address: zx@cqu.edu.cn (X. Zhao).

Table 1

Comparative summary of the reviewed studies on CC—OPWF.

Ref.	Uncertainty			Modelling		Regulation		Tractable approximation	Method for chance constraints	DNN
	Power loads	Water loads	RE	Power losses	Branch limits	Pipe direction	Pump speed			
[22]	×	×	✓	✓	✓	×	✓	Piecewise linearization	PEP	×
[23]	×	✓	×	✓	×	×	×	×	Scenario approach	×
[24]	✓	×	×	×	×	×	×	Convex hull	Scenario approach	×
[25]	✓	✓	×	×	×	×	×	Convex hull	Scenario approach	×
[26]	✓	×	×	×	×	✓	×	Quasi-convex hull	Reformulation	×
[27]	✓	✓	✓	×	✓	✓	×	Quasi-convex hull	Reformulation	×
[28]	✓	✓	×	×	✓	✓	✓	×	Fixed-point-iteration	×
This work	✓	✓	✓	✓	✓	✓	✓	SOC relaxation and McCormick envelope	Fixed-point-iteration	✓

*Ref. [28] is our previous work.

operating constraints in both systems. It also manages interdependencies between power and water. The OPWF provides an essential tool for the operation and planning of IPWSs, attracting significant research attention.

However, most of the existing OPWF models are deterministic in nature. This paper reports on the study of OPWF modeling under uncertainties, with a particular focus on an algorithm facilitated by deep neural networks (DNNs).

1.2. Literature review

1.2.1. OPWF models and methods

Many efforts have been made to address the OPWF problem at the distribution level, as outlined in [13–16]. Ref. [13] is an early contribution to OPWF formulation, where the PDN is depicted using the bus injection model, and the WDN is characterized by pipelines with predetermined directions and variable-speed pumps. The Darcy–Weisbach equations are employed to articulate the head losses in pipelines. Later investigations, as indicated in [14–16], consistently apply the branch flow model for the PDN, presupposing unfixed pipeline directions. Ref. [14] and [16] apply the Darcy–Weisbach equations, whereas [8] adopts the Hazen–Williams equations for pipeline description. In terms of pump speed control, variable-speed pumps are investigated in [15] and [9], whereas [14] is dedicated to fixed-speed pumps.

The OPWF problem is inherently a non-convex and non-linear programming problem. To address this problem, a variety of approaches have been employed, including piecewise linearization [15], second-order cone (SOC) relaxation [14,16], the convex hull technique [14], McCormick envelopes [16], and the feasible point pursuit-successive convex approximation method [13].

In addition to the OPWF problems at the distribution level, there are also studies on OPWF involving bulk transmission networks. For instance, Refs. [3] and [17] have explored the potential of water distribution networks (WDNs) in providing energy flexibility [3] or accommodating wind power [17] to bulk transmission systems through OPWF.

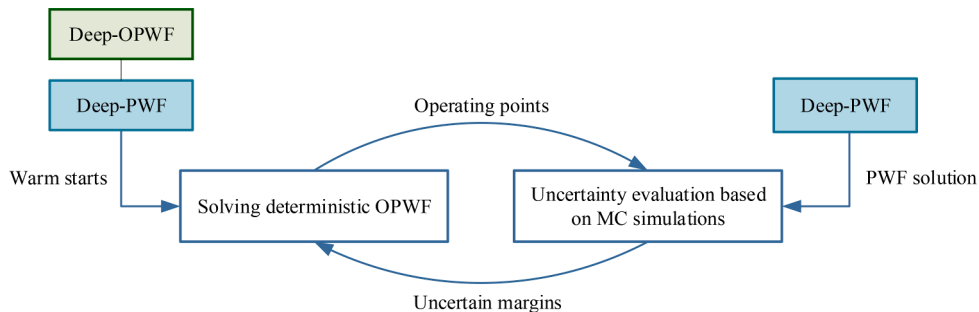
1.2.2. Chance-constrained OPWF

All the studies in [3,13–17] are deterministic optimizations, neglecting the inherent uncertainty of RE sources and load demands. However, these uncertainties can significantly affect the decision-making for OPWF. If not properly addressed, they could compromise the optimality and even the feasibility of OPWF solutions.

Common methods for addressing uncertainties in optimizations include *stochastic optimization* and *robust optimization*. Stochastic optimization typically involves modeling uncertainties as random variables with known probability distributions and dealing with expected values and probabilistic constraints [18]. In contrast, robust optimization seeks solutions that are feasible across all uncertainty scenarios, emphasizing the worst-case scenario [19]. It operates without assuming known distributions, however, robust solutions can be overly conservative. Both methods are widely used in power and energy system optimization [6, 20]. Additionally, there are hybrid methods of stochastic and robust optimization, such as distributionally robust optimization [19] or the stochastic-robust model proposed in [21] to optimize the operation of isolated energy communities.

In this paper, we focus on chance-constrained programming (CCP), which is a specific form of stochastic optimization. It introduces additional chance constraints to deterministic optimization, ensuring that the probability of violating certain constraints remains below a specified threshold.

CCP has been extensively studied in the OPF formulation in power systems. Recently, several studies have also resorted to CCP to address the OPWF problem under uncertainty. Ref. [22] formulated an OPWF problem incorporating uncertain RE generation, modeling the RE generation as a CCP problem and solving it using the probability-efficient point (PEP) method. The authors of [23–26] expanded their series of related studies to address uncertainties in water loads [23], power loads [24,26], or both [25] by incorporating chance constraints of violating network constraints into the distribution-level OPWF problem. They tackled these chance-constrained OPWF (CC—OPWF) problems using scenario-based methods [23–25]. However, due to the limited scalability of the scenario approach for larger problems, the authors proposed in

**Fig. 1.** Outline of the fixed-point-iteration algorithm assisted by DNNs.

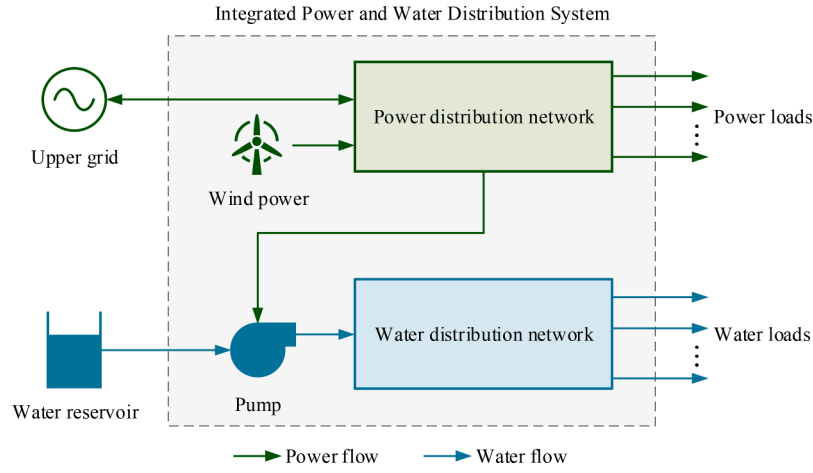


Fig. 2. Schematic diagram of the integrated system composed of a power distribution network (PDN) and a WDN.

[26] a reformulation of the OPWF problem by introducing robust water constraints and a tractable transformation of the chance constraints. A similar reformulation of chance constraints is utilized in [27] to address uncertainties in water and power loads as well as wind power. However, the reformulation outlined in [26] and [27] is based on the linearization of the inequality constraints and the assumption of Gaussian uncertainties. In our previous work [28], we introduced the *fixed-point-iteration* algorithm to address the CC—OPWF problem for the first time.

1.2.3. Fixed-point-iteration-based CC-OPF and CC-OPWF

The *fixed-point-iteration* algorithm was proposed in [29] as a solution to chance-constrained OPF. It is built upon the concept of *uncertain margin* previously defined by the same authors in [30]. The uncertain margin represents an estimation of the quantiles of uncertain variables and is interpreted as network constraint tightening to accommodate uncertainties from RE generation and power demands. The fixed-point-iteration algorithm is notable for its ability to decompose the CCP optimization process into successive steps of solving a deterministic OPF problem and conducting a probabilistic evaluation of uncertain margins. Initially, the iteration algorithm was developed based on partial linearization of the AC power flow equations as well as the assumption of Gaussian uncertainties to facilitate the analytical reformulation of chance constraints [30]. The sample-based approach, such as the Monte Carlo (MC) method, was subsequently incorporated into the iteration to accommodate non-linear models and non-Gaussian uncertainties [31]. However, the evaluation of uncertain margins typically requires tens of thousands of MC simulations at each iteration, posing a challenge to computational efficiency. To address this issue, alternative approaches such as the three-point estimation [32] and the response-surface approach [33] are employed to accelerate the uncertainty evaluation. Moreover, the non-convex nature of the OPF problem presents another challenge for the iteration algorithm, as it relies on repeatedly solving the problem. Consequently, [32] employs a linearized network model, while [34] applies SOC relaxations to address the convex problem.

To the best of our knowledge, our previous work [28] was the first and the only attempt to apply the fixed-point-iteration algorithm to CC—OPWF, however, it had limitations in network representation, intractability, and computation efficiency. Specifically, the power system was oversimplified by using the DC power model, the OPWF formulated was intractable due to nonlinear and non-convex constraints within the water network, and the probabilistic evaluation is conducted through repeatedly solving of PWF equations using the Newton-Raphson (N-R) method, posing great challenges to computation efficiency.

1.3. Research gap and contributions

From the literature review, a comparative summary of the studies on CC—OPWF is presented in Table 1. The research gaps in the context of CC—OPWF can be identified as follows:

- 1) Incomplete modeling of uncertainties, network representation, and regulations. For instance, most studies do not account for regulations concerning pipe direction and pump speed in CC—OPWF.
- 2) Challenges in effectively and efficiently solving the CC—OPWF problem. The scenario approach used in [23–25] and the fixed-point-iteration method we previously used in [28] are not applied to large-scale systems. Additionally, the reformulation methods proposed in [26] and [27] require uncertainty factors to follow Gaussian distributions and necessitate model linearization. Therefore, further research is needed to develop a more efficient and scalable method to address the intractability and computational burden associated with CC—OPWF.

To fill these identified gaps, this work aims to develop a novel method for addressing the challenges in modeling and solving the CC—OPWF problem, particularly with the assistance of deep neural networks (DNNs). Recent years have witnessed the growing successful application of DNNs in power flow [35], probabilistic power flow [36, 37], optimal power flow [38,39], optimal gas flow [40], optimal power-gas flow [41], and the integrated scheduling of power and gas systems [42]. Inspired by these applications, particularly in fitting high non-linear power flow equations [36,37] and predicting the optimal range of gas flow [41], as well as specifying *warm-starts* for power-gas optimization [40], we propose leveraging DNN to improve the fixed-point-iteration algorithm for CC—OPWF.

The main contributions of this work are twofold:

- **CC—OPWF formulation.** A CC—OPWF problem is formulated by integrating detailed power and water network constraints, bi-directional water flow, pump speed regulation, and considerations of uncertainties in power and water demands, as well as RE generations. This problem is subsequently reformulated into a tractable MISOC problem through the application of uncertain margins, SOC relaxation, and the McCormick envelopes.
- **CC—OPWF solution.** A novel fixed-point-iteration algorithm incorporating two DNNs, referred to as Deep-PWF and Deep-OPWF, is proposed to provide an efficient and scalable solution to the CC—OPWF problem. Deep-PWF is developed and trained to predict PWF solutions, thereby accelerating the MC simulations for uncertainty evaluation. The cascade of Deep-OPWF and Deep-PWF is

utilized to provide a warm start by predicting water flow directions and refining variable limits for the McCormick envelopes. The proposed iteration algorithm is outlined in Fig. 1.

The remainder of this paper is organized as follows. Section 2 describes the typical structure of the integrated distribution system under study and formulates the deterministic OPWF problem. In Section 3, the chance-constrained OPWF problem is formulated to address uncertainties and subsequently reformulated as a deterministic problem using the concept of uncertain margins. The fixed-point-iteration algorithm is also presented in this section. Section 4 presents the proposed DNNs and their incorporation into the iterative algorithm for chance-constrained OPWF. Case studies are conducted in Section 5 to validate the proposed DNNs and overall iterative algorithm. Finally, Section 6 concludes the paper and discusses future work.

2. Optimal power-water flow formulation

Fig. 2 depicts the integrated power and water distribution system studied in this work, where the power and water distribution networks are interconnected through electric pumps. These pumps consume electrical power to elevate the hydraulic heads at water sources or within the WDN, thereby compensating for head loss resulting from pipeline friction or gravity. We will first present the mathematical modeling of the integrated system shown in Fig. 2 before we formulate the optimal power-water flow problem.

2.1. Modeling of the power distribution network

The PDN studied is radial and represented as a directed graph (N_P, E_P), where N_P is the node set and E_P is the edge set representing distribution lines and transforms. The notations used throughout this article are summarized in Table A2 for convenience to look up.

We use the *relaxed branch flow model* [43] to model the radial PDN. For each branch $(i, j) \in E_P$, let $r_{ij} + jx_{ij}$ be the series impedance and $P_{ij} + jQ_{ij}$ be the sending-end complex power from node i to j , let $P_i + jQ_i$ be the complex power injection at node $i \in N_P$, we have

$$P_{ij} - r_{ij}l_{ij} + P_j = \sum_{k:j \rightarrow k} P_{jk}, \forall (i, j) \in E_P \quad (1a)$$

$$Q_{ij} - x_{ij}l_{ij} + Q_j = \sum_{k:j \rightarrow k} Q_{jk}, \forall (i, j) \in E_P \quad (1b)$$

$$v_i - v_j = 2(r_{ij}P_{ij} + x_{ij}Q_{ij}) - (r_{ij}^2 + x_{ij}^2)l_{ij}, \forall (i, j) \in E_P \quad (1c)$$

$$P_{ij}^2 + Q_{ij}^2 = v_i l_{ij}, \forall (i, j) \in E_P \quad (1d)$$

where l_{ij} is the squared current magnitude on branch (i, j) , v_i is the squared voltage magnitude at node i . Constraints (1a) and (1b) describe the nodal balance of active and reactive power at node $j \in N_P$; (1c) is the Ohm's law; (1d) defines the sending-end power flow in the branch $(i, j) \in E_P$.

For the studied PDN, which is connected to WDNs through electrical pumps, the active and reactive power injection at node i are given as:

$$P_i = P_{i,gen} + P_{i,wind} - P_{i,load} - P_{i,pump}, \forall i \in N_P \quad (2a)$$

$$Q_i = Q_{i,gen} + Q_{i,wind} - Q_{i,load}, \forall i \in N_P \quad (2b)$$

where $P_{i,gen}$, $P_{i,wind}$, $P_{i,load}$, and $P_{i,pump}$ are the active power of generators, wind turbines, load, and the electric pumps at node $i \in N_P$, respectively, while $Q_{i,gen}$, $Q_{i,wind}$, and $Q_{i,load}$ are their corresponding reactive powers.

In addition to (1a)–(1d), the following constraints on powers and magnitudes of voltage and current are required:

$$P_{i,gen}^{min} \leq P_{i,gen} \leq P_{i,gen}^{max}, \forall i \in N_P \quad (3a)$$

$$Q_{i,gen}^{min} \leq Q_{i,gen} \leq Q_{i,gen}^{max}, \forall i \in N_P \quad (3b)$$

$$v_i^{min} \leq v_i \leq v_i^{max}, \forall i \in N_P \quad (3c)$$

$$l_{ij} \leq l_{ij}^{max}, \forall (i, j) \in E_P \quad (3d)$$

$$P_{ij}^{min} \leq P_{ij} \leq P_{ij}^{max}, \forall (i, j) \in E_P \quad (3e)$$

$$Q_{ij}^{min} \leq Q_{ij} \leq Q_{ij}^{max}, \forall (i, j) \in E_P \quad (3f)$$

where the superscripts max and min are used to denote the upper and lower limits on the corresponding variables.

Note that the nodes without generators, loads, wind turbines, or pumps can be handled by simply setting the respective powers in (2a), (2b) and their upper and lower limits in (3a), (3b) to zero.

2.2. Modeling of the water distribution network

Consider a WDN composed of junctions, pipes, pumps, reservoirs, and tanks. Similar to PDN, we use another directed graph (N_W, E_W) with node set N_W and edge set E_W to describe the WDN. The node set N_W represents junctions, water reservoirs, and tanks, and the edge set E_W represents pipes and pumps as $E_W = E_{Pi} \cup E_{Pu}$.

The WDN is modeled by the following functions on pipes, pumps, and nodal flow balance:

$$H_m - H_n = \text{sgn}(H_m, H_n) f_{mn} q_{mn}^2, \forall (m, n) \in E_{Pi} \quad (4a)$$

$$\text{sgn}(H_m, H_n) = \begin{cases} 1, & H_m \geq H_n \\ -1, & H_m < H_n \end{cases}, \forall (m, n) \in E_{Pi} \quad (4b)$$

$$H_n - H_m = a_{mn} \omega_{mn}^2 - b_{mn} q_{mn}^2, \forall (m, n) \in E_{Pu} \quad (4c)$$

$$P_{mn,pump} = \frac{\rho g}{\eta_{mn}} q_{mn} (H_n - H_m), \forall (m, n) \in E_{Pu} \quad (4d)$$

$$q_{m,src} - q_{m,load} + \sum_{k:k \rightarrow m} q_{km} = \sum_{n:m \rightarrow n} q_{mn}, \forall m \in N_W \quad (4e)$$

Constraint (4a) is the *Darcy-Weisbach* equation, where H_m is the hydraulic head at node $m \in N_W$, and q_{mn} and f_{mn} are the volumetric flow rate and friction factor of pipe $(m, n) \in E_{Pi}$, respectively. Constraints (4c) and (4d) formulate the hydraulic characteristic of the pump $(m, n) \in E_{Pu}$ and its power consumption with the nonnegative coefficients of a_{mn} and b_{mn} , efficiency η_{mn} , relative speed ω_{mn} , water density ρ , and the gravity constant g [44]. Constraint (4e) represents conservation of mass requiring the total inflow to equal total outflow at each node $m \in N_W$ with the water source flow rate $q_{m,src}$ and the load flow rate $q_{m,load}$.

Note that $P_{mn,pump}$ in (4d) denotes the electric power consumed by pump $(m, n) \in E_{Pu}$. Assuming that the pump is powered by node $i \in N_P$ in the PDN, $P_{mn,pump}$ corresponds to $P_{i,pump}$ in (2a). As a result, the PDN and WDN are coupled through (2a) and (4d).

Finally, we impose the following limits on flow rates, heads, and pump speeds:

$$q_{mn}^{min} \leq q_{mn} \leq q_{mn}^{max}, \forall (m, n) \in E_W \quad (5a)$$

$$q_{m,src}^{min} \leq q_{m,src} \leq q_{m,src}^{max}, \forall m \in N_W \quad (5b)$$

$$H_m^{min} \leq H_m \leq H_m^{max}, \forall m \in N_W \quad (5c)$$

$$H_{mn}^{min} \leq H_n - H_m \leq H_{mn}^{max}, \forall (m, n) \in E_{Pu} \quad (5d)$$

$$\omega_{mn}^{min} \leq \omega_{mn} \leq \omega_{mn}^{max}, \forall (m, n) \in E_{Pu} \quad (5e)$$

$$H_{\text{ref}} = \text{const.} \quad (5f)$$

where $H_{mn}^{\min}$ and $H_{mn}^{\max}$ are the lower and upper bound of head gain through the pump $(m, n) \in E_{pu}$. Constraint (5f) states that one of the reservoirs is chosen as the reference node with a given hydraulic head.

2.3. Formulation of the optimal power-water flow problem

The objective of the OPWF problem is to minimize the overall costs associated with sourcing electricity from the upper grid and water from the reservoir, as depicted in Fig. 2. Additionally, it aims to reduce active power losses within the PDN. To achieve this goal, the following objective function has been formulated:

$$C = \sum_{i \in \mathcal{I}_p} c_{i,\text{gen}} P_{i,\text{gen}} + \sum_{m \in \mathcal{N}_w} c_{m,\text{src}} q_{m,\text{src}} + \sum_{(i,j) \in \mathcal{E}_p} r_{ij} l_{ij} \quad (6)$$

where $c_{i,\text{gen}}$ for $i \in \mathcal{N}_p$ and $c_{m,\text{src}}$ for $m \in \mathcal{N}_w$ denote the electricity and water price, respectively. The third term represents the active power loss in the PDN.

Let $\mathbf{u} = \{\mathbf{P}_{\text{gen}}, \mathbf{Q}_{\text{gen}}, \mathbf{v}_{\text{slack}}, \boldsymbol{\omega}, \mathbf{q}_{\text{src}}\}$ be the set of *control variables*, where $\mathbf{P}_{\text{gen}}$ and $\mathbf{Q}_{\text{gen}}$ are vectors representing the active and reactive generation powers at each node, respectively; $\mathbf{v}_{\text{slack}}$ denotes the squared voltage of the root node connected to the upper grid; $\boldsymbol{\omega}$ is the vector of pump speeds, and $\mathbf{q}_{\text{src}}$ aggregates all source flow rates excluding the reference reservoir. Let $\mathbf{x} = \{\mathbf{P}, \mathbf{Q}, \mathbf{v}, \mathbf{l}, \mathbf{q}, \mathbf{H}\}$ be the set of *dependent variables*, where $\mathbf{P}, \mathbf{Q}$, and $\mathbf{l}$ are vectors of P_{ij}, Q_{ij} , and $l_{ij}, (i, j) \in E_p$; $\mathbf{v}$ is the vector of $v_b, i \in \mathcal{N}_p$; $\mathbf{q}$ is the vector of $q_{mn}, (m, n) \in E_w$, and $\mathbf{H}$ is the vector of $H_m, m \in \mathcal{N}_w$. The OPWF problem can be formulated as follows:

$$\text{OPWF} : \min_{\mathbf{x}, \mathbf{u}} C(\mathbf{x}, \mathbf{u}) \quad (7a)$$

$$\text{subject to } \mathbf{f}(\mathbf{x}, \mathbf{u}) = \mathbf{0} \quad (7b)$$

$$\mathbf{g}(\mathbf{x}, \mathbf{u}) \leq \mathbf{0} \quad (7c)$$

where $C(\cdot)$ is the objective function defined by (6). Equality constraint (7b) represents the power Eqs. (1a)–(1d), (2a), (2b), and hydraulic Eqs. (4a)–(4e), (5f), which also defines the PWF problem of the integrated system. Inequality constraint (7c) collects the constraints of (3a)–(3f) and (5a)–(5e).

The OPWF poses a non-convex and non-linear challenge, utilizing sign functions to denote pipe directions. These sign functions can be effectively handled through the big-M method [16], transforming the problem into a mixed-integer nonlinear programming (MINLP) problem.

3. Chance-constrained OPWF formulation and its fixed-point-iteration solution

3.1. Uncertainty modeling

The OPWF problem above is a deterministic problem neglecting the inherent uncertainty associated with the power- and water-loads and wind powers. To address the uncertainty, the active and reactive power loads, and the flow rates of water loads are modeled as the sum of the forecasted values and forecast errors [31]:

$$\tilde{P}_{i,\text{load}} = P_{i,\text{load}} + \Delta \tilde{P}_{i,\text{load}}, \forall i \in \mathcal{N}_p \quad (8a)$$

$$\tilde{Q}_{i,\text{load}} = Q_{i,\text{load}} + \Delta \tilde{Q}_{i,\text{load}}, \forall i \in \mathcal{N}_p \quad (8b)$$

$$\tilde{q}_{m,\text{load}} = q_{m,\text{load}} + \Delta \tilde{q}_{m,\text{load}}, \forall m \in \mathcal{N}_w \quad (8c)$$

where the tilde $\sim$ is used to denote uncertain variables, and the forecast errors $\Delta \tilde{P}_{i,\text{load}}, \Delta \tilde{Q}_{i,\text{load}}$, and $\Delta \tilde{q}_{m,\text{load}}$ are all assumed to be Gaussian variables with zero means.

We follow [32] to model the wind speed v_{wind} as a two-parameter

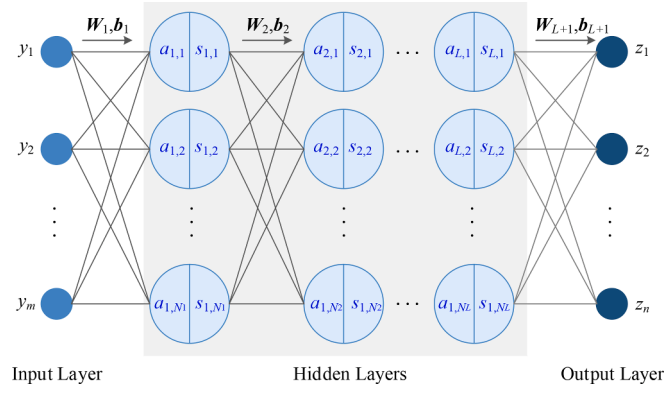


Fig. 3. Structure of the Deep-PWF, where $a_{i,j}$ and $s_{i,j}$ denote respectively the outputs of the linear mapping and the activation function of the j -th neuron in the i th layer. Assuming there are L hidden layers and N_i neurons in the i th layer, we have $i \in \{1, 2, \dots, L\}$ and $j \in \{1, 2, \dots, N_i\}$.

Weibull random with scale parameter α and shape parameter β . The cumulative distribution function is

$$F(\tilde{v}_{\text{wind}}) = 1 - \exp \left[- \left(\frac{\tilde{v}_{\text{wind}}}{\alpha} \right)^\beta \right]. \quad (9)$$

The corresponding wind power at node i can be expressed as $\tilde{P}_{\text{wind}} = f(\tilde{v}_{\text{wind}})$ where the node index is dropped to simplify the notation. We used the following function (10) to map wind speed to wind power [45]:

$$f(v) = \begin{cases} 0, & v \leq v_{\text{in}} \text{ or } v > v_{\text{out}} \\ \frac{1}{2} \rho_{\text{air}} A C_p v^3, & v_{\text{in}} < v < v_N \\ P_{\text{wind},N}, & v_N \leq v \leq v_{\text{out}} \end{cases}, \quad (10)$$

where $v_{\text{in}}, v_{\text{out}}$, and v_N are the cut-in, cut-out, and nominal wind speed, respectively; $P_{\text{wind},N}$ is the nominal wind power; ρ_{air} is the air density, A is the area swept out by the wind turbine blades, and C_p is the wind-turbine power coefficient which is regarded as constant in this work.

Assuming that the wind turbine is operating under a constant power factor $\cos \varphi$, the reactive wind power is given by

$$\tilde{Q}_{\text{wind}} = \tilde{P}_{\text{wind}} \frac{\sqrt{1 - \cos^2 \varphi}}{\cos \varphi}. \quad (11)$$

It is worth noting that actual load forecast errors and wind speeds are not necessarily expected to follow Gaussian and Weibull distributions. The proposed approach in this paper does not rely on any specific assumptions about probability distributions. Moreover, if there is sufficient real-world data available on loads and wind speeds, it can be effectively incorporated into our approach.

3.2. Chance-constrained OPWF

With the uncertainty in power and water load as well as in wind speed, the power and water flowing throughout the integrated system are also uncertain. Therefore, it is usually impossible to hold the inequality constraints of the OPWF. By introducing the chance constraints to evaluate the probabilities of holding the inequalities, we formulate the OPWF as the following chance-constrained OPWF:

$$\text{CC-OPWF} : \min \mathbb{E} \left(\sum_{i \in \mathcal{I}_p} c_{i,\text{gen}} P_{i,\text{gen}} + \sum_{m \in \mathcal{N}_w} c_{m,\text{src}} q_{m,\text{src}} + \sum_{(i,j) \in \mathcal{E}_p} r_{ij} \tilde{l}_{ij} \right) \quad (12a)$$

$$\text{subject to } \mathbf{f}(\tilde{\mathbf{x}}, \mathbf{u}) = \mathbf{0} \quad (12b)$$

$$(3a), (3b), (5b), (5e), (5f),$$

$$\mathbb{P}\{\tilde{v}_i \leq v_i^{\max}\} \geq 1 - \varepsilon_v, \forall i \in \mathcal{N}_p \quad (12c)$$

$$\mathbb{P}\{\tilde{v}_i \geq v_i^{\min}\} \geq 1 - \varepsilon_v, \forall i \in \mathcal{N}_p \quad (12d)$$

$$\mathbb{P}\{\tilde{l}_{ij} \leq l_{ij}^{\max}\} \geq 1 - \varepsilon_l, \forall (i,j) \in \mathcal{E}_p \quad (12e)$$

$$\mathbb{P}\{\tilde{p}_{ij} \leq p_{ij}^{\max}\} \geq 1 - \varepsilon_p, \forall (i,j) \in \mathcal{E}_p \quad (12f)$$

$$\mathbb{P}\{\tilde{p}_{ij} \geq p_{ij}^{\min}\} \geq 1 - \varepsilon_p, \forall (i,j) \in \mathcal{E}_p \quad (12g)$$

$$\mathbb{P}\{\tilde{q}_{ij} \leq q_{ij}^{\max}\} \geq 1 - \varepsilon_q, \forall (i,j) \in \mathcal{E}_p \quad (12h)$$

$$\mathbb{P}\{\tilde{q}_{ij} \geq q_{ij}^{\min}\} \geq 1 - \varepsilon_q, \forall (i,j) \in \mathcal{E}_p \quad (12i)$$

$$\mathbb{P}\{\tilde{q}_{mn} \leq q_{mn}^{\max}\} \geq 1 - \varepsilon_q, \forall (m,n) \in \mathcal{E}_w \quad (12j)$$

$$\mathbb{P}\{\tilde{q}_{mn} \geq q_{mn}^{\min}\} \geq 1 - \varepsilon_q, \forall (m,n) \in \mathcal{E}_w \quad (12k)$$

$$\mathbb{P}\{\tilde{H}_m \leq H_m^{\max}\} \geq 1 - \varepsilon_H, \forall m \in \mathcal{N}_w \quad (12l)$$

$$\mathbb{P}\{\tilde{H}_m \geq H_m^{\min}\} \geq 1 - \varepsilon_H, \forall m \in \mathcal{N}_w \quad (12m)$$

$$\mathbb{P}\{(\tilde{H}_n - \tilde{H}_m) \leq H_{mn}^{\max}\} \geq 1 - \varepsilon_H, \forall (m,n) \in \mathcal{E}_{pu} \quad (12n)$$

$$\mathbb{P}\{(\tilde{H}_n - \tilde{H}_m) \geq H_{mn}^{\min}\} \geq 1 - \varepsilon_H, \forall (m,n) \in \mathcal{E}_{pu} \quad (12o)$$

where E denotes expectation, and P is the probability measure. Specifically, we have the following statements.

- 1) The objective (12a) of the CC—OPWF is to minimize the expected cost due to the uncertainty in currents.
- 2) The equality constraint (12b) is the stochastic counterpart of (7b) with uncertainties considered.
- 3) The inequality constraints on generations (3a), (3b), water sources (5b), (5f), and pump speeds (5e) are considered as hard constraints, which are considered impossible to violate. The remaining inequality constraints are soft and formulated as individual chance constraints with acceptable violation probabilities of ε_v , ε_b , ε_p , ε_q , ε_q , and ε_H .
- 4) For simplicity, we follow [31] to formulate the uncertain OPWF problem with individual chance constraints to limit the violation probabilities of each constraint. However, the solution framework proposed in this work can be readily extended to the CC—OPWF with joint chance constraints.

3.3. Reformulation of the chance-constrained OPWF

The CC—OPWF problem is computationally intractable with an expected cost objective and chance constraints. To address this issue, we reformulate the problem by introducing the concept of *uncertain margin*, as detailed in [31].

We use the nodal head constraints in (12l) and (12 m) for node m as illustrative examples. Assume that the actual nodal head is uncertain

due to the stochastic variations in wind speeds and load demands and can be modeled as a random variable. This variable can be further represented as the sum of the forecasted operating point (or the scheduled value) H_m and a stochastic deviation $\Delta\tilde{H}_m$ [30–32]:

$$\tilde{H}_m = H_m + \Delta\tilde{H}_m. \quad (13)$$

Suppose the cumulative distribution function (CDF) of $\Delta\tilde{H}_m$ is $F_{Hm}(\cdot)$ and its inverse function is $F_{Hm}^{-1}(\cdot)$. Based on the definition of the CDF and its inherent monotonicity, the chance constraints of (12l) and (12 m) can be reformulated into the following expression:

$$\begin{aligned} \mathbb{P}\{\Delta\tilde{H}_m \leq H_m^{\max} - H_m\} &= F_{Hm}(H_m^{\max} - H_m) \geq 1 - \varepsilon_H \Rightarrow H_m^{\max} - H_m \\ &\geq F_{Hm}^{-1}(1 - \varepsilon_H), \end{aligned} \quad (14a)$$

$$\begin{aligned} \mathbb{P}\{\Delta\tilde{H}_m \geq H_m^{\min} - H_m\} &= 1 - F_{Hm}(H_m^{\min} - H_m) \geq 1 - \varepsilon_H \Rightarrow H_m^{\min} - H_m \\ &\leq F_{Hm}^{-1}(\varepsilon_H), \end{aligned} \quad (14b)$$

or as

$$H_m^{\min} - F_{Hm}^{-1}(\varepsilon_H) \leq H_m \leq H_m^{\max} - F_{Hm}^{-1}(1 - \varepsilon_H). \quad (15)$$

Denoting

$$\lambda_{Hm}^{\text{up}} = F_{Hm}^{-1}(1 - \varepsilon_H) \quad (16a)$$

and

$$\lambda_{Hm}^{\text{low}} = -F_{Hm}^{-1}(\varepsilon_H), \quad (16b)$$

we have

$$H_m^{\min} + \lambda_{Hm}^{\text{low}} \leq H_m \leq H_m^{\max} - \lambda_{Hm}^{\text{up}}. \quad (17)$$

It is observed that 1) λ_{Hm}^{up} , which is referred to as the *upper uncertainty margin*, actually represents the $(1 - \varepsilon_H)$ -quantile of $\Delta\tilde{H}_m$. Correspondingly, $\lambda_{Hm}^{\text{low}}$, known as the *lower uncertainty margin*, is the negative of the ε_H -quantile of $\Delta\tilde{H}_m$. 2) The chance constraints have now been reformulated into deterministic constraints by introducing λ_{Hm}^{up} and $\lambda_{Hm}^{\text{low}}$. 3) The impacts of uncertainty on the nodal head can be expressed by tightening its upper and lower limits with the tightenings λ_{Hm}^{up} and $\lambda_{Hm}^{\text{low}}$. Additionally, in this reformulation process, no assumptions were made regarding the specific distribution of the random variables. This implies that the reformulation is capable of accommodating diverse distributions.

Applying a similar reformulation to all chance constraints, we can rewrite the CC—OPWF problem as

$$\mathbf{R} - \mathbf{CC} - \mathbf{OPWF} : \min_{\mathbf{x}, \mathbf{u}} \left(\sum_{i \in \mathcal{N}_p} c_{i,\text{gen}} p_{i,\text{gen}} + \sum_{m \in \mathcal{N}_w} c_{m,\text{src}} q_{m,\text{src}} + \sum_{(i,j) \in \mathcal{E}_p} r_{ij} l_{ij} \right) \quad (18a)$$

subject to (7b), (3a), (3b), (5b), (5e), (5f),

$$v_i^{\min} + \lambda_{vi}^{\text{low}} \leq v_i \leq v_i^{\max} - \lambda_{vi}^{\text{up}}, \forall i \in \mathcal{N}_p \quad (18b)$$

$$l_{ij} \leq l_{ij}^{\max} - \lambda_{ij}^{\text{up}}, \forall (i,j) \in \mathcal{E}_p \quad (18c)$$

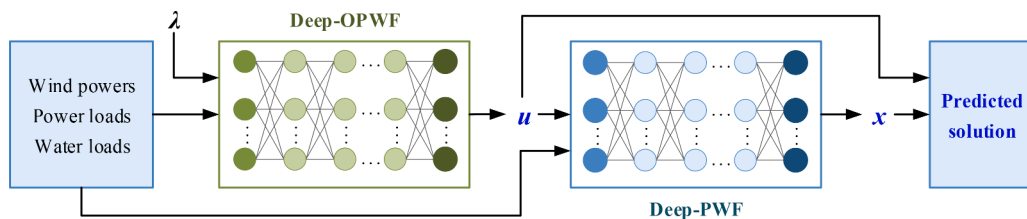


Fig. 4. Flowchart to predict the solution of the R-CC—OPWF problem.

$$P_{ij}^{\min} + \lambda_{Pij}^{\text{low}} \leq P_{ij} \leq P_{ij}^{\max} - \lambda_{Pij}^{\text{up}}, \forall (i, j) \in \mathcal{E}_P \quad (18d)$$

$$Q_{ij}^{\min} + \lambda_{Qij}^{\text{low}} \leq Q_{ij} \leq Q_{ij}^{\max} - \lambda_{Qij}^{\text{up}}, \forall (i, j) \in \mathcal{E}_P \quad (18e)$$

$$q_{mn}^{\min} + \lambda_{qmn}^{\text{low}} \leq q_{mn} \leq q_{mn}^{\max} - \lambda_{qmn}^{\text{up}}, \forall (m, n) \in \mathcal{E}_W \quad (18f)$$

$$H_m^{\min} + \lambda_{Hm}^{\text{low}} \leq H_m \leq H_m^{\max} - \lambda_{Hm}^{\text{up}}, \forall m \in \mathcal{N}_W \quad (18g)$$

$$\lambda_{Hmn}^{\text{low}} \leq H_m - H_n \leq H_m^{\max} - \lambda_{Hmn}^{\text{up}}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (18h)$$

where the upper uncertainty margins $\lambda_{vi}^{\text{up}}, \lambda_{lij}^{\text{up}}, \lambda_{Pij}^{\text{up}}, \lambda_{Qij}^{\text{up}}, \lambda_{qmn}^{\text{up}}$, and $\lambda_{Hmn}^{\text{up}}$ have similar expressions as λ_{Hm}^{up} , and the lower uncertainty margins $\lambda_{vi}^{\text{low}}, \lambda_{lij}^{\text{low}}, \lambda_{Pij}^{\text{low}}, \lambda_{Qij}^{\text{low}}, \lambda_{qmn}^{\text{low}}$, and $\lambda_{Hmn}^{\text{low}}$ have similar expressions as $\lambda_{Hm}^{\text{low}}$.

It is worth noting that the R-CC-OPWF remains a nonlinear and non-convex problem. Additionally, if all the uncertain margins are set to 0, the R-CC-OPWF reduces to an OPWF problem. Consequently, the OPWF can be considered a special instance of the R-CC-OPWF.

3.4. Fixed-point-iteration algorithm for R-CC-OPWF

We apply the fixed-point-iteration algorithm from [31] to solve the CC-OPWF problem. The algorithm alternates between solving the deterministic R-CC-OPWF problem under the given vector of tightenings $\lambda^{(k-1)}$ to obtain the optimal $[\mathbf{x}^{(k)}, \mathbf{u}^{(k)}]$, and evaluating $\lambda^{(k)}$ based on $[\mathbf{x}, \mathbf{u}]^{(k)}$ through either analytical methods or sample-based methods such as MC simulations, where k is the iteration count. The process terminates when the tightenings λ do not change between iterations.

Details of the fixed-point-iteration algorithm to solve the CC-OPWF problem are provided in Algorithm 1. MC simulation has clear advantages over analytical methods since it requires neither simplification (such as linearization) of PWF equations nor any particular assumption about the probability distribution of uncertainties. Results of MC simulations are also typically used as a reference to verify the analytical methods. However, MC simulation is usually computationally intensive. Consequently, in this work, we apply MC simulation to evaluate λ and utilize the DNN technique to mitigate the computational burden.

Algorithm 1: Fixed-point-iteration algorithm for CC-OPWF

1. **Initialize** the iteration count $k=1$ and uncertain margins $\lambda^{(k-1)}=0$.
2. **While** $k < k_{\max}$ **do**
3. **Solve** the deterministic R-CC-OPWF problem with $\lambda^{(k-1)}$ to obtain the optimal control variables $\mathbf{u}^{(k)}$ and dependent variables $\mathbf{x}^{(k)}$.

4. Evaluate $\lambda^{(k)}$ via MC simulation:

- 1) Generate samples of forecast errors in power and water loads, and wind speeds based on their probabilistic distribution.
- 2) Calculate the corresponding power and water loads and wind powers via (8a)–(8c) and (10).
- 3) Solve the PWF equations (7b) based on $\mathbf{u}^{(k)}$ and power/water loads and wind powers to obtain the samples of $\mathbf{x}^{(k)}$.
- 4) Conduct statistical analysis on the samples of $\mathbf{x}^{(k)}$ by verifying (12c)–(12o) to detect the chance constraints that do not meet the predefined violation probabilities of $\varepsilon_v, \varepsilon_b, \varepsilon_p, \varepsilon_Q, \varepsilon_q$, and ε_H .
- 5) Update the uncertain margins λ of the violated chance constraints via (16a) and (16b).
5. **Check convergence** by comparing the maximum change $|\lambda^{(k)} - \lambda^{(k-1)}|$ with the predefined threshold η . If $|\lambda^{(k)} - \lambda^{(k-1)}| \leq \eta$, go to step 7 to stop the iteration; otherwise update $k=k+1$.
6. **end while**
7. **Output** the final $\mathbf{u}^{(k)}$ and $\mathbf{x}^{(k)}$.

4. The proposed DNN-Assisted solution for CC-OPWF

Although the fixed-point-iteration algorithm successfully transforms the original CC-OPWF into alternating between solving the deterministic R-CC-OPWF and updating the uncertain margin λ , there are still two challenges. First, evaluating λ based on MC simulation is computationally intensive since it requires repeatedly solving the PWF equations (7b) multiple times. The second challenge is how to solve the nonlinear and nonconvex R-CC-OPWF problem.

To address the first challenge, we propose training a DNN-based surrogate model, referred to as Deep-PWF, to map the power and water loads, wind speeds, and control variables $\mathbf{u}$ to the dependent variables $\mathbf{x}$. As for the second challenge, we suggest reformulating the R-CC-OPWF problem into a tractable problem with SOC relaxation and McCormick envelopes. Additionally, we recommend utilizing another deep neural network, referred to as Deep-OPWF, to predict the optimal 1) nodal heads and 2) flow rates of pipes and pumps, and then use these predictions as warm starts to facilitate the solve of the R-CC-OPWF problem.

4.1. Deep-PWF

The PWF equations (7b) are nonlinear and have no closed-form solutions, which are typically solved by iterative numerical methods such

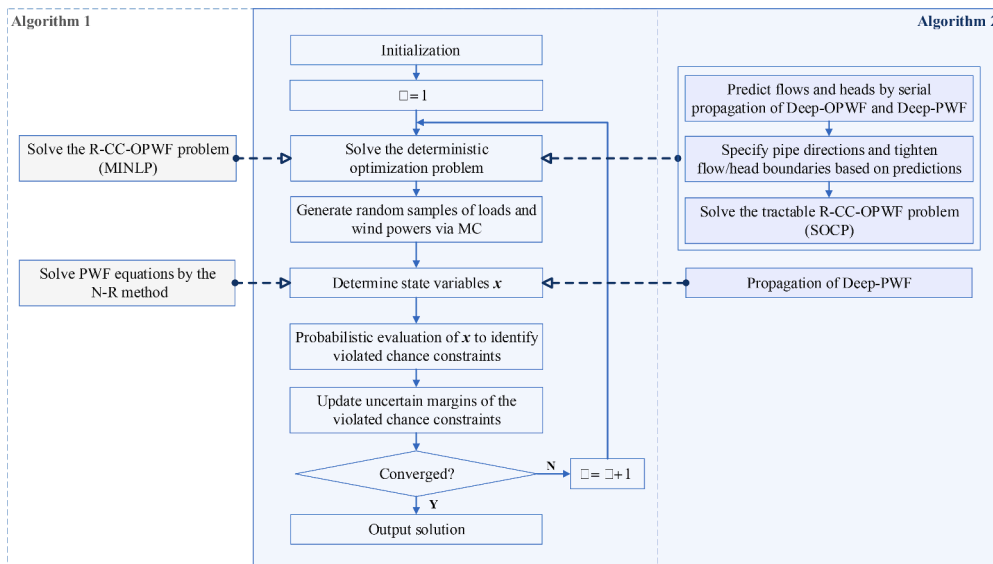


Fig. 5. Flowcharts of the two algorithms.

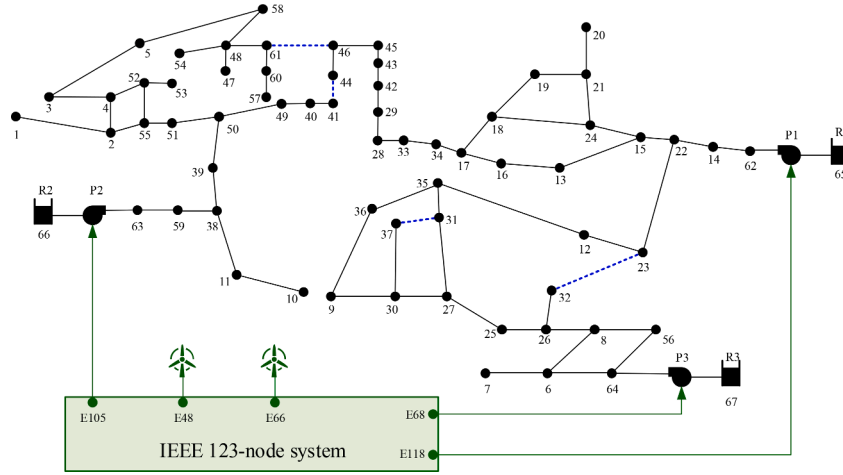


Fig. 6. Schematic of the integrated power and water distribution system. The IEEE 123-node system is coupled with the 67-node water distribution network through pumps P1, P2, and P3, which are powered by electrical nodes E118, E68, and E105 respectively. Two wind turbines with a nominal power of 39 kW each are connected to electrical nodes E48 and E66. The four blue pipes are not included in the base case C1 but are added to the extended case C2.

Table 2

Structures and training data of Deep-PWF and Deep-OPWF.

DNNs	Structure	Training			
	Layers and neurons	Inputs	Outputs	Scenarios	Simulation methods
Deep-PWF	(250, 250, 250)*	1) Power and water loads 2) Wind powers 3) Control variables u	Dependent variables x	50,000	Solve the PWF problem (7b) via the N-R method
Deep-OPWF	(500, 500, 500, 500)	1) Power and water loads 2) Wind powers 3) Uncertain margins λ	Control variables u	10,000	Solve the R-CC—OPWF problem using the solver SCIP

* (250,250,250) denotes three hidden layers with 250 neurons in each layer; (500, 500, 500, 500) follows a similar pattern.

as the well-known *Newton-Raphson* (N-R) method. However, the iteration solution itself is time-consuming, not to mention that evaluating the uncertain margins λ via MC simulations requires repeating the iteration solution of the PWF equations thousands of times or even more.

Fortunately, when applying MC simulations to repeatedly solve the PWF equations, the network topology as well as the network parameters such as impedances and pipe frictions remain unchanged. As a result, the PWF equations can be interpreted by the following mapping:

$$M_{PWF} : y_{PWF} \rightarrow z_{PWF} \quad (19)$$

where $y_{PWF} = [y_1, y_2, \dots, y_m]$ is an m -dimensional input vector collecting wind speeds, power and water loads, and control variables u , and $z_{PWF} = [z_1, z_2, \dots, z_n]$ represents the n variables involved in the chance constraints (12c)–(12o), i.e., v_b , l_{ij} , P_{ij} , Q_{ij} , H_m , and q_{mn} , where $i \in N_P \setminus \{\text{slack}\}$, $m \in N_W \setminus \{\text{ref}\}$, $(i, j) \in E_P$, and $(m, n) \in E_W$.

We suggest using a deep neural network Deep-PWF to learn the mapping (19). The Deep-PWF proposed is a multi-layer feed-forward

neural network with multiple hidden layers between the input and output layers. The structure of the Deep-PWF is depicted in Fig. 3, where each neuron is associated with 1) a linear mapping defined by a set of weights and a bias, and 2) an activation function. Accordingly, the forward propagation of the Deep-PWF is

$$s_0 = y \quad (20a)$$

$$a_k = W_k s_{k-1} + b_k, \forall k \in \{1, 2, \dots, L\} \quad (20b)$$

$$s_k = \max(a_k, 0), \forall k \in \{1, 2, \dots, L\} \quad (20c)$$

$$z = W_{L+1} s_L + b_{L+1} \quad (20d)$$

where (20a) defines the input layer, (20b) is the linear mapping, (20c) represents the *Rectifier linear unit* (ReLU) activation function selected in this work, and finally (20d) obtains the output.

The training and testing for Deep-PWF are outlined as follows:

Table 3

Mean values and STDs obtained by the Deep-PWF and N-R methods under varying load forecast error STDs and an MC size of 10,000.

Items	Methods	10 %				15 %				20 %			
		V_{68} (p.u.)	I_{12} (p.u.)	H_{10} (m)	q_{12} (m ³ /h)	V_{68} (p.u.)	I_{12} (p.u.)	H_{10} (m)	q_{12} (m ³ /h)	V_{68} (p.u.)	I_{12} (p.u.)	H_{10} (m)	q_{12} (m ³ /h)
Means	N-R	1.094	0.149	69.995	20.833	1.094	0.1484	69.995	20.832	1.094	0.149	69.995	20.831
	Deep-PWF	1.094	0.149	69.996	20.836	1.094	0.1485	69.996	20.836	1.094	0.149	69.997	20.838
	$\varepsilon\%$ *	2.8e-3	6.3e-2	2.0e-3	1.4e-2	2.6e-3	6.1e-2	1.7e-3	1.9e-2	2.8e-3	6.7e-2	2.8e-3	3.4e-2
STDs	N-R	4.1e-4	9.7e-4	0.037	2.0872	6.2e-4	0.0015	0.057	3.1174	8.5e-4	0.0020	0.074	4.1402
	Deep-PWF	4.1e-4	9.7e-4	0.037	2.0874	6.2e-4	0.0015	0.057	3.1171	8.5e-4	0.0020	0.074	4.1405
	$\varepsilon\%$	4.2e-3	8.2e-3	5.1e-3	9.6e-3	4.3e-3	3.8e-2	3.9e-3	1.4e-2	4.1e-3	4.2e-2	6.8e-3	8.2e-3

* $\varepsilon\%$ is the relative error of Deep-PWF compared to N-R; V_{68} is the voltage of node 68, I_{12} is the current of line 1–2, H_{10} is the hydraulic head of node 10, and q_{12} is the water flow rate of pipe 1–2.

- 1) Data acquisition. Due to the lack of the historical power and water flow data of the integrated system studied, we generate training data by simulation. More specifically, input data—wind speeds, power and water loads, and control variables $\mathbf{u}=\{\mathbf{P}_{\text{gen}}, \mathbf{Q}_{\text{gen}}, \mathbf{v}_{\text{slack}}, \boldsymbol{\omega}, \mathbf{q}_{\text{src}}\}$ —are randomly generated. The corresponding output data are derived by solving the PWF equations (7b) via the N-R method. All data are then divided into training and testing sets based on a specified ratio, such as 4:1.
- 2) Preprocess. We apply the *z-score* normalization method [46] to preprocess both input and output data. This method standardizes each feature by subtracting the mean and dividing it by the *standard deviation* (STD), resulting in a distribution with a mean of 0 and an STD of 1. This preprocessing not only equalizes the scales of different features but also reduces the influence of outliers, thereby facilitating a more stable and effective training phase.
- 3) Activation function choice. The widely recognized ReLU activation function is chosen due to its effectiveness in addressing the vanishing gradient problem and its computational efficiency.
- 4) Loss function definition. The *mean squared error* (MSE) is used as the loss function to evaluate the performance of the Deep-PWF. MSE provides a clear and straightforward measure of the average squared difference between the predicted and actual values, which guides the optimization process toward minimizing prediction errors.
- 5) Learning algorithm and rate adjustment: The *root mean square propagation* (RMSProp) algorithm [47] is chosen as the back-propagation algorithm. We initiate the learning rate at 0.001 and incorporate a rate reduction strategy that diminishes the learning rate by half when the loss function shows a decrease of less than a specified threshold over 5 consecutive epochs, ensuring that fine adjustments are made as the network approaches convergence.
- 6) Early stopping mechanism: To safeguard against overfitting and to promote the development of a generalizable model, an early stopping strategy is integrated into our training routine. This technique intervenes if the validation loss does not improve by more than a set threshold for 20 consecutive epochs, indicating that the model has potentially begun to memorize the training data rather than learning to generalize from it.

To improve the prediction accuracy, we employed an approach that utilizes multiple neural networks to perform the mapping of Deep-PWF, with each network learning a subset of output variables. For instance, in the *Case Studies* section, two neural networks with the same structures are trained individually to predict the power system variables of v_b , l_{ij} , P_{ij} , and Q_{ij} , and the hydraulic variables of H_m and q_{mn} .

With the trained Deep-PWF, the forward propagation process (20a)–(20d) will serve as the surrogate model for the PWF equations (7b) and consequently, the solving of the PWF problem will be significantly accelerated.

4.2. Deep-OPWF

Similar to representing the PWF problem with the mapping (19), the

R-CC—OPWF problem can also be expressed through another mapping:

$$\mathcal{M}_{\text{OPWF}} : \mathbf{y}_{\text{OPWF}} \rightarrow \mathbf{z}_{\text{OPWF}} \quad (21)$$

The difference between $\mathcal{M}_{\text{PWF}}$ (19) and $\mathcal{M}_{\text{OPWF}}$ (21) lies in the definitions of inputs $\mathbf{y}$ and outputs $\mathbf{z}$. In the newly defined $\mathcal{M}_{\text{OPWF}}$, $\mathbf{y}_{\text{OPWF}}$ collects the variables of wind speeds, power loads, water loads, as well as uncertain margins λ , while $\mathbf{z}_{\text{OPWF}}$ corresponds to the optimal solution of the R-CC—OPWF problem.

From (21), the Deep-OPWF with a structure similar to the Deep-PWF shown in Fig. 3 can be constructed and trained. We also use ReLU activation, the *z-score*-based pre-process, the RMSProp algorithm, and the MES loss function to train the Deep-OPWF. However, more hidden layers and more neurons in each hidden layer are needed in Deep-OPWF than in Deep-PWF because the mapping $\mathcal{M}_{\text{PWF}}$ is essentially the PWF equations with a relatively clear relationship between $\mathbf{y}_{\text{PWF}}$ and $\mathbf{z}_{\text{PWF}}$, while the relationship between $\mathbf{y}_{\text{OPWF}}$ and $\mathbf{z}_{\text{OPWF}}$ of the mapping $\mathcal{M}_{\text{OPWF}}$ is more complex. Moreover, the data for training Deep-OPWF is generated by solving the R-CC—OPWF problem under massive wind and load scenarios with the solver SCIP [48].

To speed up the training and guarantee satisfying the equality constraints (7b) of the R-CC—OPWF, we only use the neural networks to learn control variables, rather than simultaneously learning both control variables and state variables. That is, only $\mathbf{u}$ rather than $[\mathbf{u}, \mathbf{x}]$ is selected as the output of the Deep-OPWF. After the Deep-OPWF has been trained, it can now be used to predict $\mathbf{u}$ given $\mathbf{y}_{\text{OPWF}}$, and $\mathbf{x}$ can then be determined via solving the PWF equations (7b). Recalling that the Deep-PWF has already been trained to solve the PWF problem, the optimal solution of the R-CC—OPWF is predicted through a cascade of the two neural networks, as depicted in Fig. 4.

Although the optimal solution predicted in Fig. 4 satisfies the equality constraints of the R-CC—OPWF, it does not necessarily ensure all the inequalities are satisfied. Therefore, the predicted solution is not directly applied to the fixed-point iteration, but rather as warm-starts for the R-CC—OPWF problem. In addition, since the R-CC—OPWF is intractable due to the nonlinear power equations (1d), as well as the hydraulic Eqs. (4a)–(4d), we will apply the SOC relaxation and McCormick envelopes to approximate these constraints before using the learned warm-starts to help solve the problem.

4.3. Convex approximations utilizing warm starts from Deep-PWF and Deep-OPWF

To address the intractability of the R-CC—OPWF problem, the following convex approximation methods are applied:

- 1) The nonconvex quadratic power flow Eq. (1d) is relaxed to a convex SOC [16].
- 2) The *Darcy-Weisbach* equation is approximated by a set of linear constraints, SOCs, and McCormick envelopes by introducing two binary variables, I_{mn}^+ and I_{mn}^- , to represent the water flow directions and two sets of auxiliary variables, χ_{mn} and γ_{mn} , for each pipe [16,49].

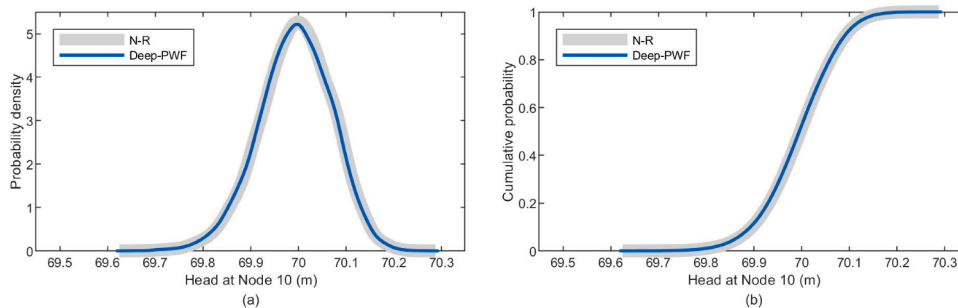


Fig. 7. (a) Probability density and (b) cumulative distribution of the head at node 10, with an STD of 10 % in load forecast error.

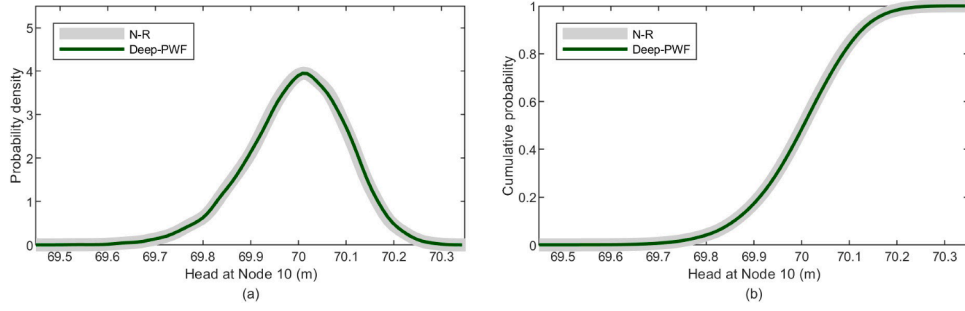


Fig. 8. (a) Probability density and (b) cumulative distribution of the head at node 10, with an STD of 20 % in load forecast error.

Table 4

Computation times for the Deep-PWF and the N-R method with different MC sample sizes and an STD of 10 % in load forecast error.

Methods	MC samples		
	5000	10,000	50,000
N-R	471.53s	963.41s	4417.35s
Deep-PWF	1.68s	2.12s	4.25s
Speed-up	280.65	454.44	1039.38

- 3) The pump [equation \(4c\)](#) is reformulated by defining $\varpi_{mn} = \omega_{mn}^2$ and replacing q_{mn}^2 with the auxiliary variable v_{mn} and its McCormick envelopes.
- 4) The pump power [Eq. \(4d\)](#) is reformulated similarly by replacing the bilinear term $q_{mn}(H_m - H_n)$ with a new auxiliary variable ζ_{mn} and its McCormick envelopes [16].

Details on the convex approximations are given in [Appendix B](#). With these convex approximations, the R-CC—OPWF problem is reformulated as a tractable mixed-integer SOC problem (MISOCP). If the water flow direction can be pre-determined, the MISOCP is further simplified to a continuous SOCP by replacing the mixed-integer constraints (A2)–(A11) with (A12)–(A16). In addition, it can be observed that the McCormick envelopes (A11), (A16), and (A22)–(A25) are closely related to the bounds of the bilinear variables, which have been proven to be critical contributors to the quality of relaxations [50]. Therefore, we are motivated to use Deep-OPWF to obtain warm-starts for two purposes: 1) identifying the water flow directions, thereby converting the original R-CC—OPWF into a tractable SOCP; and 2) tightening the variable bounds of the McCormick envelopes to improve the approximation accuracy.

Let $[q^0, H^0]$ be the predicted flow rates and nodal heads through the propagation shown in [Fig. 4](#). The following steps are used to determine the warm-starts:

- 1) Identify the water flow directions based on H_m^0 and H_n^0 via (4b) and reverse the direction with negative flows to make $I_{mn}^+ = 1$ for all $(m, n) \in E_{pi}$.
- 2) The boundaries for the McCormick envelopes (A16) and (A22)–(A25) are tightened by

$$\hat{q}_{mn}^{\min} = \max\{(1 - \alpha)q_{mn}^0, q_{mn}^{\min}\}, \hat{q}_{mn}^{\max} = \min\{(1 + \alpha)q_{mn}^0, q_{mn}^{\max}\}, \forall (m, n) \in \mathcal{E}_W \quad (22a)$$

$$\hat{H}_{mn}^{\min} = \max\{(1 - \alpha)H_{mn}^0, H_{mn}^{\min}\}, \hat{H}_{mn}^{\max} = \min\{(1 + \alpha)H_{mn}^0, H_{mn}^{\max}\}, \forall (m, n) \in \mathcal{E}_{pu} \quad (22b)$$

where $0 < \alpha < 1$ is a given constant.

4.4. CC—OPWF solution assisted by Deep-PWF and Deep-OPWF

The proposed DNN-assisted solution for CC—OPWF is summarized as a two-stage algorithm (Algorithm 2).

Algorithm 2: Fixed-point-solution for CC—OPWF assisted by DNNs
Stage 1 (Offline): Train Deep-PWF and Deep-OPWF.

1. Generate massive scenarios randomly by sampling power and water loads, wind speeds, and uncertain margins.
2. Construct inputs y_{PWF} in (19) and y_{OPWF} in (21) for training the two DNNs based on the scenarios generated.
3. Obtain outputs z_{PWF} by solving the PWF problem (7b) based on y_{PWF} for each scenario via the N-R method.
4. Obtain outputs z_{OPWF} by solving the R-CC—OPWF problem based on y_{OPWF} for each scenario with the solver SCIP.
5. Train Deep-PWF and Deep-OPWF using the massive inputs and outputs generated with the ReLU activation, the z-score-based pre-process, the RMSProp algorithm, and the MSE loss function.

Stage 2 (Online): Solve the CC—OPWF for a new scenario with the trained Deep-PWF and Deep-OPWF.

6. Initialize the iteration count $k=1$ and uncertain margins $\lambda^{(k-1)}=0$.
7. While $k < k_{\max}$ do
8. Find warm-starts for solving R-CC—OPWF:
 - 1) Predict $[q^0, H^0]$ using Deep-OPWF and Deep-PWF through the propagation shown in [Fig. 4](#);
 - 2) Specify water flow directions by (4b) and bounding by (22a) and (22b);
 - 3) Update $q_{mn}^{\min} \leftarrow \hat{q}_{mn}^{\min}, q_{mn}^{\max} \leftarrow \hat{q}_{mn}^{\max}, \forall (m, n) \in E_W$, and $H_{mn}^{\max} \leftarrow \hat{H}_{mn}^{\max}, H_{mn}^{\min} \leftarrow \hat{H}_{mn}^{\min}, \forall (m, n) \in E_{pi}$.
9. Solve the tractable R-CC—OPWF problem in [Appendix B](#) with

Table 5

Methods for comparison.

Methods	Problems	Description	Solvers
M0	R-CC—OPWF in Section 3.3	MINLP	SCIP
M1	Tractable R-CC—OPWF in Appendix B	MISOCP	Gurobi
M2 (proposed)	Tractable R-CC—OPWF in Appendix B	SOCP with warm-starts by the two DNNs	Gurobi

*M0 is referred to as the benchmark. The warm-starts are obtained through the propagation of Deep-OPWF and Deep-PWF given in [Fig. 4](#).

Table 6

Optimal results and computation time of different methods for solving the R-CC—OPWF problem of C1 and C2.

Cases	Methods	Cost (CNY)	Cost gap δ (%)	Max violation ξ	Time (s)	Speed-up
C1	M0	6433.71	—	—	5.03	—
	M1	6359.81	1.15	0.47	3.03	1.66
	M2 (proposed)	6432.80	0.01	7.45e-4	1.64	3.07
C2	M0	6430.86	—	—	23.36	—
	M1	6353.67	1.20	0.63	4.61	5.07
	M2 (proposed)	6429.95	0.01	4.54e-3	2.66	8.78

* M0 is referred to as the benchmark for calculating the cost gap and speed-up of M1 and M2.

$\lambda^{(k-1)}$ and the updated boundings to obtain $u^{(k)}$ and $x^{(k)}$.

10. Evaluate $\lambda^{(k)}$ via MC simulation:

1) Generate samples of forecast errors in power and water loads, and wind speeds based on their probabilistic distribution.

2) Calculate the corresponding power and water loads and wind powers via (8a)–(8c) and (10).

3) Obtain the samples of $x^{(k)}$ based on $u^{(k)}$ and power/water loads and wind powers through the propagation of Deep-PWF (20a)–(20d).

4) Conduct statistical analysis on the samples of $x^{(k)}$ by verifying (12c)–(12o) to detect the chance constraints that do not meet the predefined violation probabilities of ε_v , ε_b , ε_p , ε_Q , ε_q , and ε_H .

5) Update the uncertain margins λ of the violated chance constraints via (16a) and (16b).

11. Check convergence by comparing the maximum change $|\lambda^{(k)} - \lambda^{(k-1)}|$ with the predefined threshold η . If $|\lambda^{(k)} - \lambda^{(k-1)}| \leq \eta$, go to step 7 to stop the iteration; otherwise update $k = k + 1$.

12. end while

13. Output the final $u^{(k)}$ and $x^{(k)}$.

To provide a clearer illustration of Algorithms 1 and 2, flowcharts for both are presented in Fig. 5. The two algorithms share a similar iterative frame, with the main differences being in the approach to solving the deterministic optimization problem and obtaining system state variables. Specifically, Algorithm 2 introduces the following modifications: 1) It substitutes the N-R method for solving the PWF equations with the propagation process of Deep-PWF. 2) Rather than directly tackling the MINLP problem (R-CC—OPWF), Algorithm 2 adopts a sequential approach: it first employs Deep-OPWF and Deep-PWF to predict flows and heads, then specifies pipe directions and narrows variable ranges, and finally, it solves the tractable SOCP problem.

Table 7

Optimal results and computation time of the two algorithms.

Cases	Methods	Iterations	Costs (CNY)	Cost gap δ (%)	Time (s)	Speed-up
C1	Algorithm 1	3	6555.73	—	2819.4	—
	Algorithm 2 (proposed)	3	6553.96	0.027	10.6	266.0
C2	Algorithm 1	3	6530.65	—	2950.5	—
	Algorithm 2 (proposed)	3	6528.76	0.029	14.7	200.7

* Algorithm 1 is used as the reference for calculating the cost gap and the speed-up of Algorithm 2.

5. Case studies

5.1. Simulation setup

For case studies, we use the single-phase version of the IEEE 123-node feeder coupled with the 67-node water distribution network, as shown in Fig. 6. There are 3 reservoirs, 3 variable-speed pumps, and 67 pipes in the water network for the base case C1. In the extended case C2, 4 additional pipes are added as indicated by the blue dashed lines in Fig. 6. Electricity is priced at CNY 0.48 per kWh, while water from reservoirs R1, R2, and R3 is priced at CNY 0.253, 0.283, and 0.318 per m³, respectively. Comprehensive parameters for the water network and the two wind turbines are detailed in the supplementary online data file [51].

All optimal models have been implemented in YALMIP [52]. The original OPWF problem and R-CC—OPWF problem are solved using SCIP, while the tractable OPWF and R-CC—OPWF problems are addressed with Gurobi [53]. Python 3.7 and Tensorflow 2.0 [54] are utilized for constructing and training the Deep-PWF and Deep-OPWF. Table 2 provides a comparative analysis of the network structures and training datasets for the two DNNs. All the simulations are conducted on a laptop computer equipped with a 3.11 GHz Intel Core i7–9700H processor and 16 GB RAM.

It is worth noting that the mapping of Deep-OPWF is more complex than Deep-PWF, requiring more hidden layers and neurons. However, Deep-PWF undergoes a larger-scale sample training compared to Deep-OPWF. This is because Deep-PWF is used directly to evaluate uncertainty margins, while Deep-OPWF only provides warm-starts for solving the R-CC—OPWF problem. Additionally, due to the similar topologies of C1 and C2, transfer learning [55] was applied to enhance the training efficiency of the two DNNs in C2.

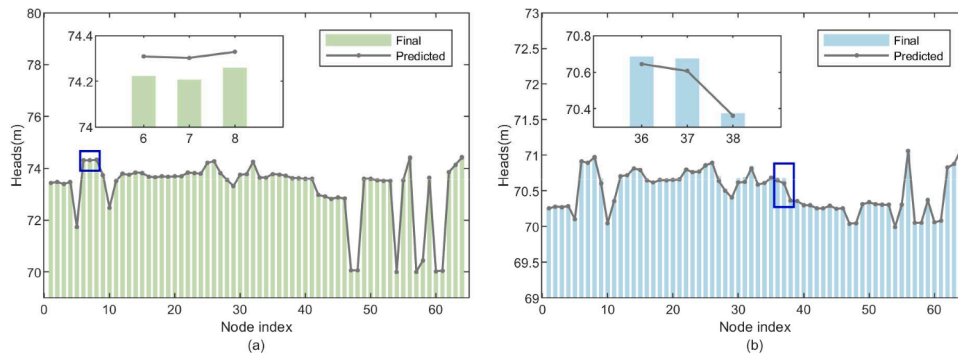


Fig. 9. Predicated heads of non-reservoir nodes (i.e., Node 1~Node 64) by the DNNs and the final optimal heads for (a) C1 and (b) C2. The predicated heads are from the propagation of Deep-OPWF and Deep-PWF given in Fig. 4.

Table 8

Maximum violations of chance constraints in percent for the three methods.

Cases	Models	Methods	Voltages	Branch currents	Heads	Pipe and pump flow rates
C1	Deterministic OPWF	SCIP	48.67 %	51.02 %	50.16 %	66.54 %
	Chance-constrained OPWF	Algorithm 1	4.85 %	4.86 %	4.98 %	4.33 %
	Chance-constrained OPWF	Algorithm 2	4.82 %	4.84 %	4.96 %	4.29 %
C2	Deterministic OPWF	SCIP	46.96 %	49.61 %	51.62 %	66.24 %
	Chance-constrained OPWF	Algorithm 1	4.99 %	4.32 %	4.99 %	4.66 %
	Chance-constrained OPWF	Algorithm 2	4.99 %	4.27 %	4.98 %	4.58 %

5.2. Evaluation of the Deep-PWF

As explained in Section 4.1, the trained Deep-PWF functions essentially as a surrogate model of the PWF equations (7b). Consequently, its forward propagation can be considered a deterministic solver for these equations. We evaluate its performance in this section with the following steps: 1) Generate massive samples randomly by sampling Gaussian forecast errors in loads and the Weibull random wind speeds via MC simulation. 2) Determine all the voltages, currents, power flows, nodal heads, and water flow rates through all pipes and pumps of each scenario given a set of control variables using the trained Deep-PWF and the N-R method respectively to solve (7b). 3) Evaluate the relative errors between the results of Deep-PWF and the N-R method.

Both C1 and C2 are simulated and similar evaluation results are obtained. Only the results of C2 are provided and discussed below. Let σ be the standard deviations (STDs) of the Gaussian forecast errors in power and water loads (8a)–(8c). Assuming σ is 10 %, 15 %, and 20 % of the forecasted loads. Table 3 compares the mean values and STDs of the 10,000 scenarios by MC simulation with the trained Deep-PWF and the N-R method. Due to space limitations, only four variables are presented as demonstration examples.

Table 3 shows that both means and STDs derived from the two methods are almost the same across all assessed variables and levels of load forecast errors. The maximum relative error ($\varepsilon\%$) of means and STDs does not exceed $6.7\text{e-}2$. This suggests that the Deep-PWF is a reliable surrogate for the N-R method in predicting system states. Furthermore, it is noted that with an increase in the STD of forecast error (σ), the STDs of all four variables increase accordingly, while the means remain relatively stable, as anticipated.

The probability distributions obtained by the two methods are also compared. Fig. 7 and Fig. 8 respectively present the cumulative distribution of H_{10} for σ being 10 % and 20 % of the forecasted value. The distribution curves of the Deep-PWF closely match those of the N-R method very well. Additionally, the distribution of H_{10} closely resembles a Gaussian distribution, with a mean of approximately 70 m, consistent with the simulation results in Table 3. When comparing the scenario with a 10 % STD in Fig. 7 to a 20 % STD in Fig. 8, the PDF curve in Fig. 8 appears flatter and wider, indicating a broader range of possible values for H_{10} and a more dispersed distribution as expected.

Results of Table 3, Fig. 7, and Fig. 8 validate the precision of the Deep-PWF. We also evaluate the computation efficiency of both methods in Table 4 with an STD of 10 % in load forecast error and MC sample sizes of 5000, 10,000, and 50,000. The results clearly indicate that the Deep-PWF exhibits a significant efficiency advantage over the N-R method, completing all simulations within seconds and achieving a

speed-up of over 280 for all sample sizes. Moreover, as the sample size grows, the computation time of the Deep-PWF only slightly increases, while the time required by the N-R method is nearly proportional to the number of samples. These findings indicate that the Deep-PWF model is more scalable and better suited for larger simulations.

5.3. Evaluation of the DNN-assisted optimization method

In this section, we evaluate the performance of the proposed DNN-assisted optimization method in terms of solution quality and computation time. For a random scenario of the base case C1 and extended case C2, we apply the three methods shown in Table 5 to solve the R-CC–OPWF problem.

The optimal results are summarized in Table 6, where the *cost gap* δ and *max violation* ξ are calculated respectively by

$$\delta = |\text{Cost} - \text{Cost}_{\text{M0}}| / \text{Cost}_{\text{M0}} \quad (23a)$$

$$\xi = \max \left\{ \left| \frac{H_m - H_n}{f_{mn} q_{mn}^2} - 1 \right|, \forall (m, n) \in \mathcal{E}_{\text{pi}}, \left| \frac{H_n - H_m}{a_{mn} \omega_{mn}^2 - b_{mn} q_{mn}^2} - 1 \right|, \left| \frac{\eta_{mn} P_{mn, \text{pump}}}{\rho g q_{mn} (H_n - H_m)} - 1 \right|, \forall (m, n) \in \mathcal{E}_{\text{pu}} \right\}. \quad (23b)$$

The following observations can be drawn from Table 6:

- 1) In comparison to the base case C1, the extended case C2 shows a slight decrease in cost but a significant increase in computation time for all three methods. This is attributed to the integration of 4 additional pipelines in C2, leading to an increase in the loop count within the WDN while the load flow rate on each node remains unchanged.
- 2) The cost gap for M1 is 1.15 % in C1 and 1.20 % in C2, suggesting a minor cost increase compared to M0. As for M2, the cost gap is negligible at 0.01 % for both cases, indicating that M2 achieves a cost comparable to the benchmark method.
- 3) The maximum violation for M1 is 0.47 in C1 and 0.63 in C2, which is relatively significant. Conversely, M2 shows a much better performance with a maximum violation of $7.45\text{e-}4$ in C1 and $4.54\text{e-}3$ in C2.
- 4) M1 reduces the computation time to 3.03 s in C1 and 4.61 s in C2, resulting in speed-ups of 1.66 and 5.07 times, respectively. Furthermore, M2 enhances the computation efficiency further, achieving speed-ups of 3.07 and 8.78 times in each case. This significant reduction by M2 highlights its superiority in computational efficiency. Additionally, when tackling the extended case C2 with more loops, M2 experiences the smallest increase in solution time

Table 9

Optimal results of Algorithm 2 for solving the CC–OPWF model with different acceptable violation probabilities.

Cases	Costs of OPWF (CNY)	$\varepsilon=1\%$				$\varepsilon=5\%$				$\varepsilon=10\%$			
		Costs (CNY)	ε_{max} (%)	Iter.	Time (s)	Costs (CNY)	ε_{max} (%)	Iter.	Time (s)	Costs (CNY)	ε_{max} (%)	Iter.	Time (s)
C1	6433.71	6592.85	0.98	5	19.6	6553.96	4.99	3	10.6	6525.19	10.1	2	7.1
C2	6430.86	6584.16	0.99	5	26.4	6528.76	4.99	3	14.7	6513.77	9.9	2	9.2

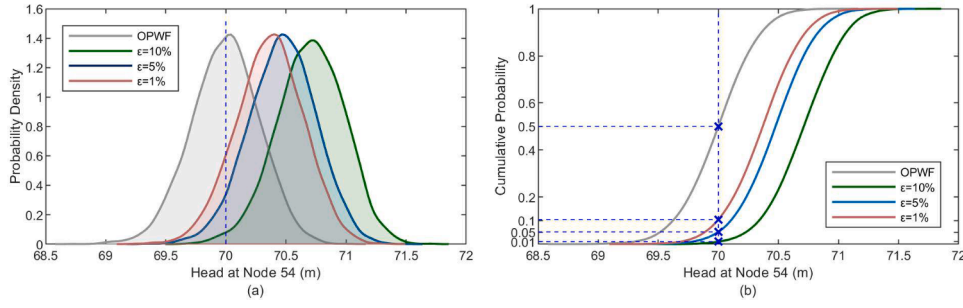


Fig. 10. (a) Probability density and (b) cumulative distribution of H_{54} with different acceptable violation probabilities. The vertical lines at 70 m represent the lower limit on the head of Node 54.

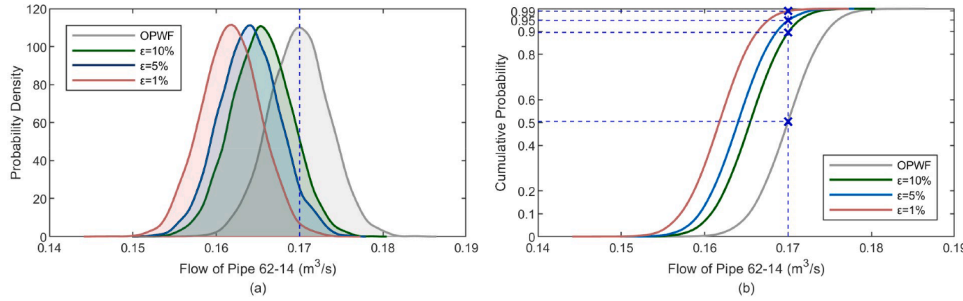


Fig. 11. (a) Probability density and (b) cumulative distribution of q_{62-14} with different acceptable violation probabilities. The vertical lines at $0.17 m^3/s$ represent the upper limit on the flow of pipe 62–14.

among the three methods, indicating good scalability for larger networks.

All these observations highlight the good performance of M2 in terms of both accuracy and efficiency and the advantages of utilizing DNNs for obtaining warm starts in solving the R-CC—OPWF problem.

Fig. 9 further compares the predicted heads by DNNs with the final optimal heads to evaluate the quality of the warm starts. The analysis shows that the predicted heads for all nodes in the base case C1 closely match the final optimal results, with a maximum difference of 0.1 m at Node 7. Similar results are observed when comparing the predicted values with the final optimal solution for C2. Fig. 9 demonstrates how DNNs provide high-quality warm starts, contributing to the improved performance of the proposed method.

5.4. Evaluation of the DNN-assisted iteration algorithm for CC-OPWF

After evaluating the performance of the Deep-PWF and Deep-OPWF separately, we finally assess the performance of the proposed iteration solution for CC—OPWF assisted by the two DNNs in this section.

1) Comparison between Algorithm 1 and Algorithm 2

We enforce all the acceptable violation probabilities for chance constraints to be 5 %, and all the STDs of the forecasted error in power and water loads are 10 % of their forecasted value. Algorithm 1 uses SCIP to solve the R-CC—OPWF, and Algorithm 2 uses Gurobi to solve the tractable problem with warm-starts given by the two DNNs. Both algorithms evaluate the uncertain margins based on 10,000 scenarios generated by MC simulation.

Results of the base case C1 and extended case C2 are listed in Table 7, where the results of Algorithm 1 are referred to as a reference for cost gap and speed-up. It can be observed that both algorithms require only 3 iterations for both cases. Algorithm 2 yields a slightly lower cost compared to Algorithm 1, but the cost gap between the two algorithms is less than 0.03 %. In terms of computation time, Algorithm 1 requires nearly 50 min to complete the calculations, while Algorithm 2 takes less than 15 s, demonstrating a significant speed advantage.

After solving the CC—OPWF problem with the two algorithms, we further compare the optimal solutions by evaluating the probabilities violating the chance constraints (see Eqs. (12c)–(12o)) based on the 10,000 scenarios generated by MC simulation. For each scenario, we solve the PWF problem (7b) with Deep-PWF based on the optimal control variables obtained by the two algorithms and record the observed probabilities violating the chance constraints. Table 8 presents the maximum violations in voltages, branch currents, heads, and flow rates of pipes and pumps. The violations in the optimal solution from the deterministic OPWF model are also included for comparison. It is evident that the deterministic OPWF, which does not account for uncertainties, results in an approximate 50 % probability of violating constraints. However, both algorithms used to solve the chance-constrained OPWF effectively maintain the violation probabilities at or below the acceptable threshold of 5 %, showcasing the effectiveness of incorporating chance constraints into the OPWF model.

2) Performance of Algorithm 2 with different acceptable violation probabilities

Algorithm 2 is applied to solve the CC—OPWF problem for C1 and C2 with different acceptable violation probabilities of 1 %, 5 %, and 10 %. The load forecasting is subjected to the same STDs of 10 %. Table 9 provides the costs, computation time, iterations, and violation probabilities, where $\epsilon_{\max}$ (%) represents the maximum probability, expressed as a percentage, of violating any chance constraint. The costs of the deterministic OPWF model are also included.

The following conclusions can be drawn from Table 9:

- 1) The operating costs of the deterministic OPWF model are consistently lower than those of the CC—OPWF in both C1 and C2. As the acceptable violation probability ϵ increases, the costs for CC—OPWF decrease as expected, reflecting a trade-off between security and economy. This trade-off allows system operators to choose an ϵ value that balances economic efficiency with the desired level of constraint violation risk.
- 2) The $\epsilon_{\max}$ (%) for CC—OPWF is well within the acceptable limits set by ϵ for all scenarios. This demonstrates the effectiveness of

Algorithm 2 in finding solutions that satisfy the predefined risk thresholds.

3) The computation time and number of iterations required by Algorithm 2 to converge increase as the acceptable violation probability decreases (from $\varepsilon=10\%$ to $\varepsilon=1\%$). This suggests that finding solutions with a lower risk of constraint violation is computationally more demanding.

Fig. 10 compares the PDF and CDF of H_{54} the head at node 54 obtained from the deterministic OPWF and CC—OPWF models to show the potential risk for H_{54} falling below the lower limit. Specifically, when using the deterministic OPWF model to determine the optimal operating point of the WDN, the mean value of H_{54} is approximately 70 m. The corresponding probability of H_{54} being below 70 m, denoted as $P\{H_{54}\leq 70\text{ m}\}$, is around 50 %. Conversely, when using the CC—OPWF model and decreasing the acceptable violation probability ε from 10 % to 5 % and 1 %, both curves in Fig. 10 shift to the right. As a result, the value of $P\{H_{54}\leq 70\text{ m}\}$ decreases accordingly from 10 % to 5 % and 1 %, indicating a reduced risk of violating the lower head limit.

Fig. 11 further compares the PDF and CDF of q_{62-14} , the flow rate at pipe 62–14, and provides the potential risk for the pipe exceeding its upper flow limit. Similar results to Fig. 10 are obtained. The deterministic OPWF model shows that the mean value of q_{62-14} is $0.17\text{m}^3/\text{s}$, and the probability $P\{q_{62-14}\geq 0.17\text{m}^3/\text{s}\}$ is around 50 %. By using the CC—OPWF model and decreasing ε from 10 % to 5 % and 1 %, both curves in Fig. 11 shift to the left, indicating a reduced risk of violating the upper flow limit.

The values of $P\{H_{54}\leq 70\text{ m}\}$ and $P\{q_{62-14}\geq 0.17\text{m}^3/\text{s}\}$ are in exact agreement with the preset limit violation probability ε . This confirms that the CC—OPWF model can accurately manage the risk level and also indicates that the uncertainty margins within the fixed-point-iteration algorithms can effectively capture the probabilistic nature of uncertainties.

6. Conclusions

We have expanded upon our previous work on CC—OPWF by reformulating a tractable problem and integrating DNNs into the fixed-point-iteration algorithm. The optimization model has been recast as a tractable MISOCP problem through SOC relaxation and McCormick envelope. Two DNNs are developed and trained to enhance the algorithm: one predicts the PWF solution to accelerate the uncertainty evaluation based on MC simulations, while the other provides warm-

starts for optimization by predicting water flow directions and refining variable limits. Numerical results from the IEEE 123-node feeder coupled with the 67-node water distribution network validate the computational accuracy and efficiency of the two DNNs and the overall iterative algorithms.

The main results are summarized as follows:

- 1) DNN accurately replicates the PWF solution and achieves over a 200-fold speed-up for uncertainty evaluation.
- 2) With DNN warm starts, the cost gap for OPWF reduces from 1.2 % to 0.01 %, and the maximum violation of relaxed equality constraints decreases from around 0.7 to below $4.5\text{e-}3$, while saving nearly 40 % of computation time.
- 3) The proposed DNN-assisted approach provides an effective balance between security and economy for power-water network coordination under uncertainty. Compared to the existing iterative algorithm, the DNN-assisted algorithm proposed achieves a relative error of only 0.03 % in operation costs, while significantly reducing the solution time with a speedup ratio exceeding 200.

In future work, we plan to expand the proposed iterative approach to the OPWF problem with joint chance constraints. Additionally, we intend to explore physics-informed learning approaches to enhance the learning capabilities of our DNNs, as suggested in [56].

CRediT authorship contribution statement

Xia Zhao: Writing – review & editing, Visualization, Methodology. **Dingyi Liu:** Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Zhenyu Wu:** Methodology, Conceptualization. **Chongqi Huang:** Writing – review & editing, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data has been uploaded to the online database, please refer to reference [51] in the manuscript for details

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.epsr.2024.110599](https://doi.org/10.1016/j.epsr.2024.110599).

Appendix A. Nomenclature

Tables A1 and A2.

Table A1

Acronyms.

Notations	Definitions	Notations	Definitions
CC—OPWF	Chance constraint optimal power and water flow	CCP	Chance-constrained programming
DNN	Deep neural networks	PEP	Probability-efficient point
OPF	Optimal power flow	MC	Monte Carlo
OPWF	Optimal power and water flow	N-R	Newton-raphson
PWF	Power and water flow	MINLP	Mixed-integer nonlinear programming
IPWS	Integrated power-water system	MISOCP	Mixed-integer second-order cone programming
WDN	Water distribution network	STD	Standard deviation
PDN	Power distribution network	PDF	Probability density function
SOC	Second-order cone	CDF	Cumulative distribution function
RE	Renewable energy		

Table A2
Notations.

Notations	Definitions
$N_p, E_p,$	Set of the nodes and edges of the power distribution network
N_w, E_w, E_{pi}, E_{pu}	Set of the nodes, edges, pipes, and pumps of the water distribution network
r_{ij}, x_{ij}	Series resistance and reactance of the branch $(i, j) \in E_p$
f_{mn}	The friction factor of pipe $(m, n) \in E_{pi}$
$a_{mn}/b_{mn}, \eta_{mn}$	Coefficients and efficiency of the pump $(m, n) \in E_{pu}$
ρ, g	Water density and the gravity constant
$c_{i,gen}$	Electricity price of the generator at node $i \in N_p$
$c_{m,src}$	Water price of the source at node $m \in N_w$
ρ_{air}	Air density
A	The area swept out by the wind turbine blades
C_p	The wind-turbine power coefficient
$v_{wind}, v_{in}, v_{out}, v_N$	Wind speed and the cut-in, cut-out, and nominal wind speed
$P_{wind,N}$	The nominal wind power
$\cos\varphi$	Power factor
$\varepsilon_v, \varepsilon_b, \varepsilon_p, \varepsilon_{Qp}, \varepsilon_{Qr}, \varepsilon_H$	Acceptable violation probability of voltage, current, active power, reactive power, flow rate, and hydraulic heads
$\lambda^{up}, \lambda^{low}$	Upper and lower uncertainty margins
P_{ij}, Q_{ij}	Sending-end active and reactive power of the branch $(i, j) \in E_p$
I^j	Squared current magnitude on branch $(i, j) \in E_p$
v_i	Squared voltage magnitude at node $i \in N_p$
P_i, Q_i	Active and reactive power injection at node $i \in N_p$
$P_{i,gen}, Q_{i,gen}$	Active and reactive power of generators at node $i \in N_p$
$P_{i,wind}, Q_{i,wind}$	Active and reactive wind power at node $i \in N_p$
$P_{i,load}, Q_{i,load}$	Active and reactive power of loads at node $i \in N_p$
$P_{i,pump}, P_{mn,pump}$	Electric power consumed by the pump $(m, n) \in E_w$ connected to node $i \in N_p$
H_m	Hydraulic head at node $m \in N_w$
q_{mn}	Volumetric flow rate in pipe $(m, n) \in E_{pi}$
ω_{mn}	Relative speed of pump $(m, n) \in E_{pu}$
$q_{m,src}, q_{m,load}$	Flow rate of sources and loads at node $m \in N_w$

*Subscripts indicating indexes (e.g., i, m) or variables (e.g., H for heads) are represented in italicized letters, while other subscripts appear in regular letters.

Appendix B. Tractable Reformulation for the OPWF and R-CC—OPWF

Both OPWF and R-CC—OPWF are intractable due to the nonlinear power equations (1d), as well as the hydraulic Eqs. (4a)–(4d). We approximate these constraints by using SOC relaxation and the McCormick envelopes for the bilinear and quadratic terms involved.

The nonconvex power flow Eq. (1d) is relaxed to a convex SOC [43]:

$$P_{ij}^2 + Q_{ij}^2 \leq v_i l_{ij}, \forall (i, j) \in \mathcal{E}_p \quad (A1)$$

The Darcy-Weisbach Eq. (4a) is converted to the following linear constraints, SOCs and McCormick envelopes by introducing 1) binary variables I_{mn}^+ , I_{mn}^- to indicate the water flow direction (i.e., $I_{mn}^+ = 1$ if and only if $q_{mn} \geq 0$, and $I_{mn}^- = 1$ if and only if $q_{mn} \leq 0$) and 2) auxiliary variables χ_{mn} and γ_{mn} [16,49]:

$$I_{mn}^+ + I_{mn}^- = 1, I_{mn}^+, I_{mn}^- \in \{0, 1\}, \forall (m, n) \in \mathcal{E}_{pi} \quad (A2)$$

$$q_{mn}^{\min}(1 - I_{mn}^+) \leq q_{mn} \leq q_{mn}^{\max}(1 - I_{mn}^-), \forall (m, n) \in \mathcal{E}_{pi} \quad (A3)$$

$$f_{mn} q_{mn}^2 \leq \chi_{mn}, \forall (m, n) \in \mathcal{E}_{pi} \quad (A4)$$

$$f_{mn} \gamma_{mn} \geq \chi_{mn}, \forall (m, n) \in \mathcal{E}_{pi} \quad (A5)$$

$$q_{mn}^2 \leq \gamma_{mn}, \forall (m, n) \in \mathcal{E}_{pi} \quad (A6)$$

$$\chi_{mn} \geq H_n - H_m - M(I_{mn}^+ - I_{mn}^- + 1), \forall (m, n) \in \mathcal{E}_{pi} \quad (A7)$$

$$\chi_{mn} \geq H_m - H_n + M(I_{mn}^+ - I_{mn}^- - 1), \forall (m, n) \in \mathcal{E}_{pi} \quad (A8)$$

$$\chi_{mn} \leq H_n - H_m + M(I_{mn}^+ - I_{mn}^- + 1), \forall (m, n) \in \mathcal{E}_{pi} \quad (A9)$$

$$\chi_{mn} \leq H_m - H_n - M(I_{mn}^+ - I_{mn}^- - 1), \forall (m, n) \in \mathcal{E}_{pi} \quad (A10)$$

$$\gamma_{mn} \leq (q_{mn}^{\min} + q_{mn}^{\max}) q_{mn} - q_{mn}^{\min} q_{mn}^{\max}, \forall (m, n) \in \mathcal{E}_{pi} \quad (A11)$$

where M is a sufficiently large number and (A3) is the variant of (5a) with direction indicators I_{mn}^+ and I_{mn}^- incorporated.

If the water flow directions for all the pipes can be pre-determined, i.e., $I_{mn}^+ = 1, I_{mn}^- = 0$, (A2)–(A11) reduced to

$$0 \leq q_{mn} \leq q_{mn}^{\max}, \forall (m, n) \in \mathcal{E}_{pi} \quad (A12)$$

$$f_{mn} q_{mn}^2 \leq H_n - H_m, \forall (m, n) \in \mathcal{E}_{pi} \quad (A13)$$

$$f_{mn}\gamma_{mn} \geq H_n - H_m, \forall (m, n) \in \mathcal{E}_{Pi} \quad (A14)$$

$$q_{mn}^2 \leq \gamma_{mn}, \forall (m, n) \in \mathcal{E}_{Pi} \quad (A15)$$

$$\gamma_{mn} \leq (q_{mn}^{\min} + q_{mn}^{\max})q_{mn} - q_{mn}^{\min}q_{mn}^{\max}, \forall (m, n) \in \mathcal{E}_{Pi} \quad (A16)$$

The pump Eq. (4c) is approximated by defining $\varpi_{mn} = \omega_{mn}^2, v_{mn} = q_{mn}^2$, and the latter one is relaxed to the following SOC and McCormick envelopes:

$$H_n - H_m = a_{mn}\varpi_{mn} - b_{mn}v_{mn}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A17)$$

$$q_{mn}^2 \leq v_{mn}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A18)$$

$$v_{mn} \leq (q_{mn}^{\min} + q_{mn}^{\max})q_{mn} - q_{mn}^{\min}q_{mn}^{\max}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A19)$$

It is worth noting that 1) ω_{mn}^2 only appears in (4c), there is, therefore, no need to convexify $\varpi_{mn} = \omega_{mn}^2$; 2) the bounds on speed ω_{mn} (5e) becomes the following bounds on squared speed:

$$\varpi_{mn}^{\min} \leq \varpi_{mn} \leq \varpi_{mn}^{\max}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A20)$$

The power consumption (4d) is approximated similarly by introducing an auxiliary variable ζ_{mn} and its McCormick envelopes to replace the bilinear term on the right-hand side:

$$P_{mn,pump} = \frac{\rho g}{\eta_{mn}} \zeta_{mn}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A21)$$

$$\zeta_{mn} \geq q_{mn}^{\min}H_{mn} + q_{mn}H_{mn}^{\min} - q_{mn}^{\min}H_{mn}^{\min}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A22)$$

$$\zeta_{mn} \geq q_{mn}^{\max}H_{mn} + q_{mn}H_{mn}^{\max} - q_{mn}^{\max}H_{mn}^{\max}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A23)$$

$$\zeta_{mn} \leq q_{mn}^{\max}H_{mn} + q_{mn}H_{mn}^{\min} - q_{mn}^{\max}H_{mn}^{\min}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A24)$$

$$\zeta_{mn} \leq q_{mn}H_{mn}^{\max} + q_{mn}^{\min}H_{mn} - q_{mn}^{\min}H_{mn}^{\max}, \forall (m, n) \in \mathcal{E}_{Pu} \quad (A25)$$

Take the R-CC—OPWF as an example. With the approximations above, the R-CC—OPWF is reformulated to the following tractable problem:

Tractable R-CC—OPWF: $\min_{x,u} \left(\sum_{i \in \mathcal{N}_p} c_{i,gen} P_{i,gen} + \sum_{m \in \mathcal{N}_w} c_{m,src} q_{m,src} + \sum_{\forall (i,j) \in \mathcal{E}_p} r_{ij} l_{ij} \right) + \sum_{(m,n) \in \mathcal{E}_{Pi}} \tau_{mn} \gamma_{mn} + \sum_{(m,n) \in \mathcal{E}_{Pu}} \psi_{mn} v_{mn}$ (A26) subject to 1) Power flow equations: (1a)–(1c), (A1), (2a), (2b);
 2) Hydraulic equations: (A2)–(A11), (A17)–(A19), (A21)–(A25);
 3) Hard inequalities: (3a), (3b), (5b), (A20), (5f)
 4) Reformulated chance constraints: (18b)–(18h). where $\tau_{mn}, (m, n) \in \mathcal{E}_{Pi}$ and $\psi_{mn}, (m, n) \in \mathcal{E}_{Pu}$ are penalty coefficients to tighten the SOCs of (A15) and (A18).

References

- [1] D. Bauer, M. Philbrick, B. Vallario, et al., The Water-Energy Nexus: Challenges and Opportunities, US Department of Energy, 2014.
- [2] S. Ahmad, H. Jia, Z. Chen, et al., Water-energy nexus and energy efficiency: a systematic analysis of urban water systems, *Renew. Sustain. Energy Rev.* 134 (2020) 110381.
- [3] K. Oikonomou, M. Parvania, Optimal coordination of water distribution energy flexibility with power systems operation, *IEEE Trans. Smart. Grid.* 10 (1) (2019) 1101–1110.
- [4] K. Smith, Understanding Energy Consumption for Urban Water Supply in China, MS thesis, Dept. Environ. Sci. Eng, Tsinghua Univ, Beijing, China, 2015.
- [5] E. Dall'Anese, P. Mancarella, A. Monti, Unlocking flexibility: integrated optimization and control of multienergy systems, *IEEE Power Energy Mag.* 15 (1) (2017) 43–52.
- [6] M. Nasir, A.R. Jordehi, M. Tostado-Véliz, et al., Two-stage stochastic-based scheduling of multi-energy microgrids with electric and hydrogen vehicles charging stations, considering transactions through pool market and bilateral contracts, *Int. J. Hydrog. Energy* 48 (61) (2023) 23459–23497.
- [7] F. Qi, M. Shahidepour, Z. Li, et al., A chance-constrained decentralized operation of multi-area integrated electricity–natural gas systems with variable wind and solar energy, *IEEE Trans. Sustain. Energy* 11 (4) (2019) 2230–2240.
- [8] Y. Meng, S.A. Mansouri, A.R. Jordehi, et al., Eco-environmental scheduling of multi-energy communities in local electricity and natural gas markets considering carbon taxes: a decentralized bi-level strategy, *J. Clean. Prod.* 440 (2024) 140902.
- [9] X. Liu, J. Wu, N. Jenkins, et al., Combined analysis of electricity and heat networks, *Appl. Energy* 162 (2016) 1238–1250.
- [10] W. Wei, S. Mei, L. Wu, et al., Optimal traffic-power flow in urban electrified transportation networks, *IEEE Trans. Smart. Grid.* 8 (1) (2016) 84–95.
- [11] A. Shabanpour-Haghighi, A.R. Seifi, An integrated steady-state operation assessment of electrical, natural gas, and district heating networks, *IEEE Trans. Power Syst.* 31 (5) (2015) 3636–3647.
- [12] Y. Jiang, Z. Ren, C. Lu, et al., A region-based low-carbon operation analysis method for integrated electricity–hydrogen–gas systems, *Appl. Energy* 355 (2024) 122230.
- [13] A.S. Zamzam, E. Dall Anese, C. Zhao, et al., Optimal water-power flow-problem: formulation and distributed optimal solution, *IEEE Trans. Control Netw. Syst.* 6 (1) (2019) 37–47.
- [14] Q. Li, S. Yu, A.S. Al-Sumaiti, et al., Micro water-energy nexus: optimal demand-side management and quasi-convex hull relaxation, *IEEE Trans. Control Netw. Syst.* 6 (4) (2019) 1313–1322.
- [15] K. Oikonomou, M. Parvania, Optimal coordinated operation of interdependent power and water distribution systems, *IEEE Trans. Smart. Grid.* 11 (6) (2020) 4784–4794.
- [16] X. Li, Z. Wu, L. Yang, et al., A convex-relaxation based method for optimal water-power flow, *Energy Rep.* 8 (2022) 973–983.
- [17] Y. Yao, C. Li, C. Shao, et al., Efficient operation of integrated electrical–water system for wind power accommodation, *IEEE Trans. Industr. Inform.* 19 (9) (2023) 9382–9393.
- [18] S.S. Reddy, V. Sandeep, C. Jung, Review of stochastic optimization methods for smart grid, *Front. Energy* 11 (2017) 197–209.
- [19] Antonio J. Conejo, Wu X, Robust optimization in power systems: a tutorial overview, *Optim. Eng.* 23 (4) (2022) 2051–2073.
- [20] D. Bertsimas, E. Litvinov, X.A. Sun, et al., Adaptive robust optimization for the security constrained unit commitment problem, *IEEE Trans. Power Syst.* 28 (1) (2012) 52–63.
- [21] M. Tostado-Véliz, A.R. Jordehi, S.A. Mansouri, et al., Day-ahead scheduling of 100% isolated communities under uncertainties through a novel stochastic-robust model, *Appl. Energy* 328 (2022) 120257.
- [22] L. Edmonds, M. Derby, M. Hill, et al., Coordinated operation of water and electricity distribution networks with variable renewable energy and distribution locational marginal pricing, *Renew. Energy* 177 (2021) 1438–1450.
- [23] A. Stuhlmacher, J.L. Mathieu, Chance-constrained water pumping managing power distribution network constraints, in: 2019 North American Power Symposium (NAPS), 2019, pp. 1–6.
- [24] A. Stuhlmacher, J.L. Mathieu, Water distribution networks as flexible loads: a chance-constrained programming approach, *Electr. Power Syst. Res.* 188 (2020) 106570.

- [25] A. Stuhlmacher, J.L. Mathieu, Chance-constrained water pumping to manage water and power demand uncertainty in distribution networks, *Proc. IEEE* 108 (9) (2020) 1640–1655.
- [26] Stuhlmacher A., Mathieu J.L., "Uncertainty-aware methods for leveraging water pumping flexibility for power networks," arXiv preprint arXiv:2207.04943, 2022.
- [27] N. Gao, C. Wang, N. Liu, et al., Chance-constrained operation of the water-energy nexus with conflicting objectives, *IET Energy Syst. Integr.* 3 (2) (2021) 184–201.
- [28] Z. Wu, M. Sun, Y. Liang, et al., Chance-constrained optimal power-water flow considering load uncertainties, in: 2021 IEEE 5th Conference on Energy Internet and Energy System Integration (EI²), 2021, pp. 4023–4028.
- [29] J. Schmidli, L. Roald, S. Chatzivasileiadis, et al., Stochastic AC optimal power flow with approximate chance-constraints, in: 2016 IEEE Power and Energy Society General Meeting (PESGM), 2016, pp. 1–5.
- [30] H. Qu, L. Roald, G. Andersson, Uncertainty Margins For Probabilistic AC Security Assessment, *IEEE*, 2015.
- [31] L. Roald, G. Andersson, Chance-constrained AC optimal power flow: reformulations and efficient algorithms, *IEEE Trans. Power Syst.* 33 (3) (2018) 2906–2918.
- [32] X. Du, X. Lin, Z. Peng, et al., Chance-constrained optimal power flow based on a linearized network model, *Int. J. Electric. Power Energy Syst.* 130 (2021) 106890.
- [33] Y. Xu, M. Korkali, L. Mili, et al., An iterative response-surface-based approach for chance-constrained AC optimal power flow considering dependent uncertainty, *IEEE Trans. Smart. Grid.* 12 (3) (2021) 2696–2707.
- [34] L. Halilbasić, P. Pinson, S. Chatzivasileiadis, Convex relaxations and approximations of chance-constrained AC-OPF problems, *IEEE Trans. Power Syst.* 34 (2) (2018) 1459–1470.
- [35] X. Hu, H. Hu, S. Verma, et al., Physics-guided deep neural networks for power flow analysis, *IEEE Trans. Power Syst.* 36 (3) (2020) 2082–2092.
- [36] Y. Yang, Z. Yang, J. Yu, et al., Fast calculation of probabilistic power flow: a model-based deep learning approach, *IEEE Trans. Smart. Grid.* 11 (3) (2019) 2235–2244.
- [37] D. Wang, K. Zheng, Q. Chen, et al., A data-driven probabilistic power flow method based on convolutional neural networks, *Int. Trans. Electric. Energy Syst.* 30 (7) (2020) e12367.
- [38] W. Huang, X. Pan, M. Chen, et al., Deepopf-v: solving ac-opf problems efficiently, *IEEE Trans. Power Syst.* 37 (1) (2021) 800–803.
- [39] R. Nellikkath, S. Chatzivasileiadis, Physics-informed neural networks for AC optimal power flow, *Electric Power Syst. Res.* (2022) 108412.
- [40] H. Liu, L. Yang, X. Shen, et al., A data-driven warm start approach for convex relaxation in optimal gas flow, *IEEE Trans. Power Syst.* 36 (6) (2021) 5948–5951.
- [41] S. Bao, Z. Yang, L. Guo, et al., One-segment linearization modeling of electricity-gas system optimization, *Energy* 197 (2020) 117230.
- [42] H. Liu, X. Shen, Q. Guo, et al., A data-driven approach towards fast economic dispatch in electricity-gas coupled systems based on artificial neural network, *Appl. Energy* 286 (2021) 116480.
- [43] M. Farivar, S.H. Low, Branch flow model: relaxations and convexification—Part I, *IEEE Trans. Power Syst.* 28 (3) (2013) 2554–2564.
- [44] Walski T.M., Chase D.V., Savic D.A., et al., "Advanced water distribution modeling and management," 2003.
- [45] F. Moazeni, J. Khazaei, Co-optimization of wastewater treatment plants interconnected with smart grids, *Appl. Energy* 298 (2021) 117150.
- [46] H. Anysz, A. Zbiciak, N. Ibadov, The influence of input data standardization method on prediction accuracy of artificial neural networks, *Procedia Eng.* 153 (2016) 66–70.
- [47] R. Setiono, H. Liu, Feature extraction via neural networks. Feature Extraction, Construction and Selection: A Data Mining Perspective, 1998, pp. 191–204.
- [48] T. Achterberg, SCIP: solving constraint integer programs, *Math. Program. Comput.* 1 (2009) 1–41.
- [49] S. Chen, A.J. Conejo, R. Sioshansi, et al., Unit commitment with an enhanced natural gas-flow model, *IEEE Trans. Power Syst.* 34 (5) (2019) 3729–3738.
- [50] H. Nagarajan, M. Lu, E. Yamangil, et al., Tightening McCormick relaxations for nonlinear programs via dynamic multivariate partitioning, in: Principles and Practice of Constraint Programming: 22nd International Conference, CP 2016, Toulouse, France, 2016, pp. 369–387. September 5–9, 2016, Proceedings 22.
- [51] D. Liu, 67 Node IEWS Data. <https://doi.org/10.13140/RG.2.2.23794.11207>.
- [52] J. Lofberg, YALMIP: a toolbox for modeling and optimization in MATLAB, in: 2004 IEEE International Conference on Robotics and Automation (IEEE Cat. No. 04CH37508), 2004, pp. 284–289.
- [53] Gurobi Optimization, The Gurobi Optimizer, available online from <http://www.gurobi.com>.
- [54] Abadi M., Agarwal A., and Barham P., et al., "Tensorflow: large-scale machine learning on heterogeneous systems," 2015, Software available from tensorflow.org, URL <https://www.tensorflow.org/>.
- [55] S.J. Pan, Q. Yang, A survey on transfer learning, *IEEE Trans. Knowl. Data Eng.* 22 (10) (2009) 1345–1359.
- [56] G.S. Misyris, A. Venzke, S. Chatzivasileiadis, Physics-informed neural networks for power systems, in: 2020 IEEE Power & Energy Society General Meeting (PESGM), 2020, pp. 1–5.